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Early Cretaceous Paleolatitude of the Yukon-Koyukuk Province, Alaska

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We report results from a paleomagnetic study of the Yukon-Koyukuk province, a key region for reconstructing the Cretaceous paleogeography of northern Alaska and northeastern Asia. The province lies between the displaced continental fragment of Arctic Alaska and the accreted terranes of southern Alaska. Although Lower Cretaceous volcanogenic rocks of the Yukon-Koyukuk province preserve primary remanent magnetizations, the primary component is masked by a pervasive overprint. The overprinting magnetization, which has dual polarity, was acquired after a Late Cretaceous orogeny left the region highly deformed. By fitting planes to tilt-corrected thermal demagnetization data and solving for the best intersection of the corresponding remagnetization circles, we separated the primary component, presumably a vestige of the original thermoremanent magnetization, from the overprinting component of the natural remanence. The primary magnetization corresponds to a mean paleopole of Early Cretaceous age at 60.2°N, 159°E. The mean paleolatitude is 69°N with the 95% confidence ellipse ranging from 60° to 75°N. This analysis was successful because the tilt corrections of the sampled strata had sufficient variation to ensure a strong intersection of the remagnetization circles. We infer that Lower Cretaceous rocks of the Brooks Range and the Yukon-Koyukuk province were once separated by a wide ocean basin that was closed by subduction, probably during the Cretaceous. The northern limb of the Angayucham terrane, which lies between the Brooks Range and the Yukon-Koyukuk province, may contain remnants of the now-closed ocean basin.

INTRODUCTION

Lower Cretaceous volcanic rocks of the Koyukuk terrane of northern Alaska have been interpreted as the products of an intraoceanic arc that probably formed on a basement of oceanic crust [*Patton*, 1973; *Box*, 1985]. The volcanic arc and the overlying Cretaceous marine sedimentary rocks make up the Yukon-Koyukuk province, which forms a wedge between continental rocks of the Brooks Range, Seward Peninsula, and the Ruby terrane (Figure 1). Bounded by ophiolites, high-pressure metamorphic rocks, and the major thrust belt of the Brooks Range, the Yukon-Koyukuk province has the geological attributes of a displaced microplate. We undertook this paleomagnetic study to test for possible displacement of the Lower Cretaceous rocks of the Koyukuk terrane, using the paleolatitude grid of stable North America as a reference frame.

Most paleomagnetic research in Alaska has concentrated on the accretionary history of the terranes that are south of the Yukon-Koyukuk province [*Stone et al.*, 1982; *Hillhouse*, 1987], and only a few studies have been attempted in the Arctic region. In the Koyukuk basin, *Harris et al.* [1987] determined paleolatitudes from mid-Cretaceous (Albian and Cenomanian) sedimentary rocks and Eocene volcanic flows. They concluded that the Cretaceous rocks originated in the northern Pacific region and have been displaced to higher latitudes relative to the North American craton by approximately 15° of latitude. Their Eocene paleolatitude determination is concordant with the cratonic reference pole, and this result constrains the timing of the inferred displacement to the interval 90–56 m.y. ago. From the Lower Cretaceous Nanushuk Group of the western North Slope, *Witte et al.* [1987] determined a paleolatitude of 75°N, again suggesting poleward displacement of about 10° from an original position in the Pacific basin. In contrast, the Lower Cretaceous Kuparuk River Formation yielded a paleopole indicating a substantial counterclockwise rotation and a shift to higher latitude as the Arctic Alaska terrane rotated out of the Canada Basin according to a paleomagnetic study by *Halgedahl* and *Jarrard* [1987], whose samples were obtained from deep exploratory boreholes near Prudhoe Bay.

In general, Cretaceous paleolatitude determinations from the sedimentary basins of northern Alaska range from 65° to 80°N, indicating deposition took place inside a radius of about 2500 km from the Cretaceous paleomagnetic axis, which is near the Bering Strait in present-day coordinates. Within this region, however, active plate tectonic forces have been at work. The continental blocks of Alaska and eastern Siberia have shifted relative to each other since the Early Cretaceous, as new ocean basins formed by rifting while older basins were closed by subduction [*Fujita*, 1978; *Fujita and Newberry*, 1982; *Harbert et al.*, 1987]. The Canada Basin, for example, probably formed during the Cretaceous as the continental fragment of Arctic Alaska rifted from the Sverdrup margin in Canada and rotated counterclockwise [*Tailleur*, 1969; *Hamilton*, 1970; *Grantz et al.*, 1979; *Halgedahl and Jarrard*, 1987]. The continental carbonate rocks of Arctic Alaska continue into Chukotka in the USSR [*Churkin and Trexler*, 1980; *Churkin et al.*, 1985], so the rotation of this continental block influenced tectonic events in northeastern Asia as well. Along the southern boundary of the Arctic Alaska–Chukotka blocks there is a major suture that contains fragments of oceanic crust, which separates the Brooks Range from the Yukon-Koyukuk province. In the northeast USSR the suture extends into the southern Anyuy belt and separates Chukotka from the Omolon continental block [*Fujita*, 1978; *Parfenov and Natal'in*, 1986].

The collision of the Koyukuk terrane with the continental platform of Arctic Alaska occurred during pre-Albian time according to geologic evidence summarized by *Box and Patton* [1988]. The timing of the collision is constrained by a distinctive belt of 110-m.y.-old alkalic plutons that crosses the suture into the Seward Peninsula and by the Albian age of overlapping sedimentary sequences. Acceptance of the paleo-

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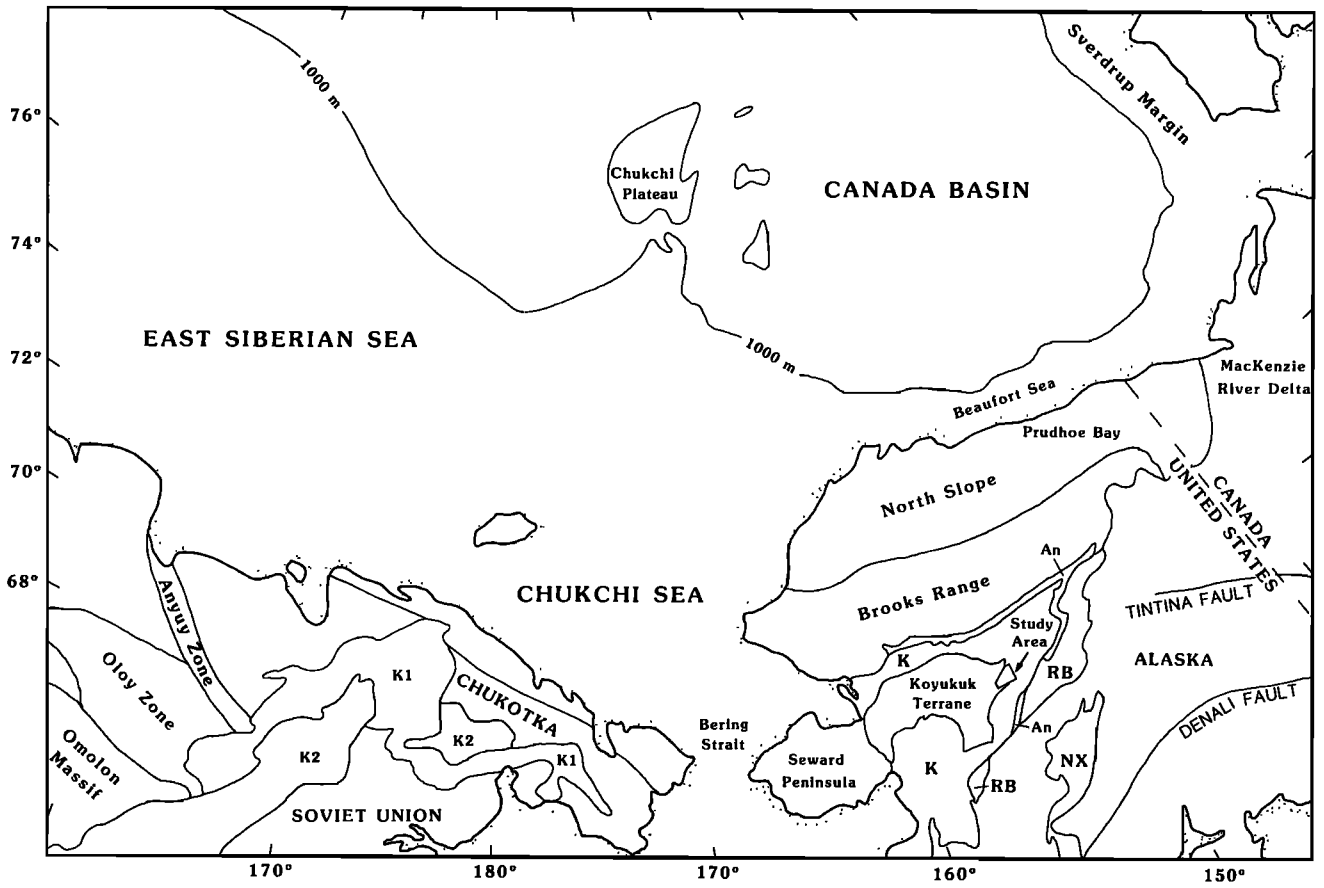


Fig. 1. Geologic provinces of Arctic Alaska and northeast USSR and location of study area (Figure 2) within the Koyukuk terrane. Associated Alaskan terranes [*Jones et al.*, 1987]: An, Angayucham; NX, Nixon Fork; RB, Ruby. K, Cretaceous sedimentary rocks of the Yukon-Koyukuk Province. In the Soviet Union, Lower Cretaceous (K1) and Upper Cretaceous (K2) rocks of the Okhotsk-Chukotsk plutonic belt.

latitude determination of *Harris et al.* [1987] from the Albian sedimentary rocks implies that the Koyukuk terrane and the continental rocks of the Brooks Range moved northward together from the northern Pacific region. This interpretation, however, is inconsistent with *Halgedahl and Jarrard*'s [1987] paleomagnetic results from the North Slope and the geological arguments for the rift opening of the Canada basin. The disagreement regarding paleolatitudes is more likely due to poor resolution of the paleomagnetic method than to a fundamental misunderstanding of the geologic evidence for the timing of the collision. Although the paleolatitude anomalies in question are significant in terms of their 95% confidence limits, which gauge the internal consistency of each data set, these limits do not include uncertainties arising from sources of systematic error, such as unremoved magnetic overprints or the flattening of magnetic inclination that occurs in some sedimentary rocks during deposition and diagenesis. At paleolatitudes above 65°, small errors in magnetic inclination are nearly doubled when transformed to the paleolatitude frame, simply due to the dipole formula. Therefore the movements of terranes that formed near the geographic pole are difficult to track with a resolution in paleolatitude of better than 10°.

To expand the paleomagnetic data base for northern Alaska, we focused our study on the Lower Cretaceous (Neocomian) volcanic rocks and Albian sedimentary rocks of the Koyukuk terrane, as designated by *Jones et al.* [1987] and *Howell et al.* [1985]. Most of our samples were taken in the volcanic sections because we feel that the thermoremanent magnetization process is intrinsically more stable than the depositional remanence that is acquired at low temperature by magnetite-bearing sedimentary rocks and because thermoremanence is not prone to systematic flattening of the magnetic inclination. This report supersedes the preliminary results reported by *Hillhouse and Gromme* [1985].

GEOLOGIC SETTING

The study area is located in the northeastern part of the Yukon-Koyukuk province near the village of Hughes (Figure 2). The better outcrops for paleomagnetic study are found in the steep banks along the Koyukuk River and its larger tributaries, such as Hughes Creek. The geology of most of the region has been covered only in reconnaissance, although the well-exposed areas along the Koyukuk River have been mapped in more detail [*Patton and Miller*, 1966; *Patton et al.*, 1978]. Discoveries of age-diagnostic fossils are concentrated in these exposures, and the structure of the bedding is more clearly defined along the river course.

The regional stratigraphy records a long history of volcanism, plutonism, and sedimentation of volcanically derived detritus. As indicated by fossils of several species of *Buchia*, the oldest rocks known in the area are Early Cretaceous (Neocomian) in age. These include nonmarine, shallow marine, and deep marine rocks, which form part of an extensive subduction-related volcanic arc in Alaska and the northeast

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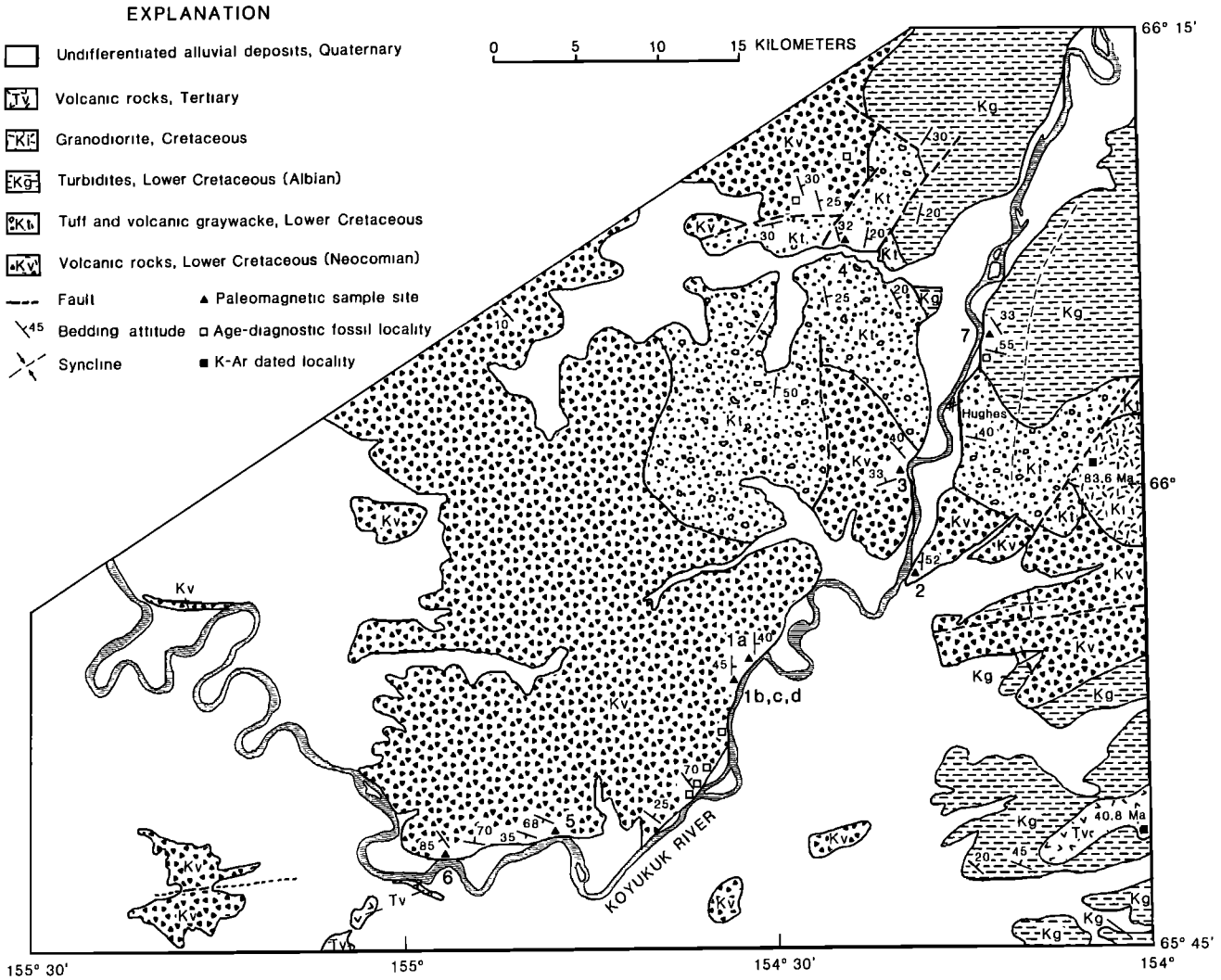


Fig. 2. Geologic map of the study area from *Patton and Miller* [1966] and *Patton et al.* [1978], showing locations of paleomagnetic samples.

USSR [*Box*, 1985]. The sequence consists mainly of flows, tuff, volcanic breccia, and hypabyssal intrusions. Chemical compositions are dominantly andesitic. Lesser amounts of tuffaceous chert, fossiliferous limestone, and pillow basalt are present. Five age determinations which were obtained from flows and intrusions by the potassium-argon method, range from 134 ± 5 Ma to 117 ± 4 Ma [*Patton and Miller*, 1966; *Patton*, 1973].

The andesitic volcanic rocks grade upward into a sequence of Hauterivian to Aptian shales and tuffs that were deposited in a deeper slope environment [*Box et al.*, 1985]. These deposits are in turn overlain by basinal turbidites of mid-Cretaceous (Albian) age. Near Hughes, the turbidites are composed of rhythmically interbedded layers of fine graywacke and mudstone. The detritus was derived in part from the Neocomian volcanic rocks and also from the late Paleozoic and early Mesozoic terranes that rim the Yukon-Koyukuk province. Ammonite fossils from the vicinity of Hughes indicate that the turbidites are late Early Cretaceous (Albian) in age. According to the Mesozoic time scale of *Harland et al.* [1982] the turbidites were deposited during approximately 97–113 Ma.

Two suites of plutonic rocks, which might have thermally affected the paleomagnetism of the Cretaceous section, occur within the eastern part of the Yukon-Koyukuk province. The older suite consists of quartz monzonite such as the Melozitna pluton dated at 111 Ma [*Patton et al.*, 1978]. These plutons mainly intrude rocks of the Ruby and Angayucham terranes and are nowhere known to intrude rocks of the Koyukuk terrane. The younger plutonic suite, which is Late Cretaceous in age, intrudes the Neocomian volcanic rocks and mid-Cretaceous turbidites of the Koyukuk terrane. In our study area the Indian Mountain pluton (dated by K-Ar at 83.6 ± 3.0 Ma) cuts folds within the Albian turbidites, indicating that the intrusion postdated deformation or occurred during its final stages [*Patton and Miller*, 1966] (age recalculated for new decay constants according to Table 2 of *Dalrymple* [1979]). The pluton is surrounded by a contact aureole of hornfels, 1–4 km wide. Numerous small dikes and sills are associated with the Late Cretaceous plutons.

Volcanic rocks of Tertiary age unconformably overlie mid-Cretaceous turbidites in the southern part of our study area. These rocks primarily consist of basalt, rhyolitic welded tuff, rhyolite flows, and breccia. Two samples of obsidian from the Little Indian River area (Figure 2) yielded a mean weighted K-Ar age of 40.8 ± 0.8 Ma [*Miller and Lanphere*, 1981].

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TABLE 1. Paleomagnetic Sample Sites in the Yukon-Koyukuk Province, Alaska

Site	Latitude °N	Longitude °E	Bedding Strike	Bedding Dip	Stratigraphic Thickness, m
Turbidities, Lower Cretaceous (Albian)					
7a	66.08	205.78	331.5°	33°E	66
7b	66.08	205.78	329°	47°E	
7c	66.08	205.78	265.8°	115°N	75
Volcanic Rocks, Lower Cretaceous (Neocomian and Barremian)					
1a	65.91	205.46	359°	40.5°E	5
1b	65.90	205.45	001°	45°E	
1c	65.90	205.45	337°	86°E	20
1d	65.90	205.45	005°	45.5°E	
2a	65.95	205.68	185°	52°W	250
2b	65.95	205.68	182°	34°W	
3	66.01	205.68	254°	33°N	50
4a	66.13	205.60	352°	32°E	30
4b	66.13	205.60	350°	32°E	
5	65.80	205.06	141.5°	85°W	5
6	65.81	205.20	112.5°	68°S	70

Harris et al. [1987] reported ages of 43.0 ± 1.3 Ma and 43.7 ± 1.3 Ma (K-Ar on whole rock) from basalt flows exposed on the south bank of the Koyukuk River.

The general structural trends in the Koyukuk River area are roughly parallel to the northeast trending southern boundary of the Yukon-Koyukuk province [*Patton*, 1973]. However, all pre-Tertiary rocks are much deformed, and more complicated structures have been superimposed on the general trends. In the area of our study the Cretaceous rocks are tightly folded and are cut by numerous faults, as indicated by steep dips and reversals of dip over short distances. Many of the structures predate the Late Cretaceous Indian Mountain pluton, and only moderate tilting is observed in the Tertiary volcanic rocks, indicating that much of the deformation occurred between 110 and 80 Ma.

PALEOMAGNETIC METHODS

Using portable drilling equipment, we collected oriented samples from the Neocomian volcanic rocks, Barremian tuffs, and Albian turbidites of the Koyukuk terrane (Figure 2 and Table 1). In the volcanic rocks we usually collected eight cores from individual cooling units, each defined as a flow, welded tuff, or intrusive body. In the volcaniclastic rocks and turbidites, each site is represented by a group of oriented cores that are distributed evenly within short stratigraphic sections ranging from a few meters thick up to 75 m thick. A total of 29 time samples (cooling units plus sections) were collected in the Neocomian and Barremian sequence; three short stratigraphic sections were collected in the Albian turbidites. Structural control was quite good at sites 1–4 and 7, where fine-grained volcaniclastic beds are well exposed. The control was less certain at site 6, which consisted of massive volcanic and intrusive units. There, a thin chert bed within the volcanic rocks was measured for the structural correction. At site 5, a single flow of pillow basalt, we relied on the pillow structure to obtain the bedding correction.

We measured all remanent magnetizations with a cryogenic magnetometer. Alternating field (af) demagnetizations were carried out in a shielded apparatus that is equipped with a self-reversing tumbler. Typically, af demagnetization treatments consisted of 13 steps in progressively higher fields up to a peak field of 100 mT. Thermal demagnetization was performed on batches of 24 specimens by heating and cooling the batch in a shielded furnace. Heatings were run in air in an ambient direct field of less than 5 nT. During the intervals between heatings, the specimens were routinely stored in a magnetically shielded compartment to reduce newly acquired viscous remanent magnetization. In doing the thermal demagnetization runs, we usually heated the specimens at 50° intervals up to 300°C, then reduced the interval to 25°. Most of the thermal runs were stopped at 575°C when the magnetization vector began to change erratically; however, some hematite-bearing specimens required additional heating steps up to 675°C.

ANALYSIS OF COMPONENTS OF REMANENT MAGNETIZATION

General Procedures

Two or more pilot cores were selected from each site for detailed demagnetization runs. Companion specimens were cut from the same oriented core, so that we could directly compare the magnetic components remaining after af and thermal demagnetization. After plotting the demagnetization results on orthogonal vector diagrams, we selected endpoints of line segments and used principal component analysis to determine secondary components and characteristic magnetizations [*Kirschvink*, 1980]. In all but a few cases we fit a least squares line through the data, not constraining the line to pass through the origin. For the remaining cases, particularly for specimens that were resistant to af demagnetization, we anchored the least squares line to the origin for the computation of characteristic magnetizations. Analysis of the pilot demagnetization runs showed that magnetic overprinting of the primary remanence is severe in the Koyukuk region. Therefore, after deciding on the most effective cleaning method for each rock unit, we subjected the remaining specimens to the same sequence of steps used in the detailed pilot demagnetization runs. Inspection of equal-area projections and orthogonal vector diagrams indicated that most specimens carried several components of magnetization with overlapping coercivity distributions in the course of the af experiments and overlapping unblocking temperature distributions during the thermal runs. We therefore analyzed demagnetiza-

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tion planes using the methods of principal component analysis of *Kirschvink* [1980] and *Onstott* [1980]. The sequence of magnetization vectors from each demagnetization run was fitted with a plane passing through the origin, then the intersection of all planes was analyzed in uncorrected and tilt-corrected coordinate systems. A synthetic example of this method is illustrated in Figure 3. To estimate the precision

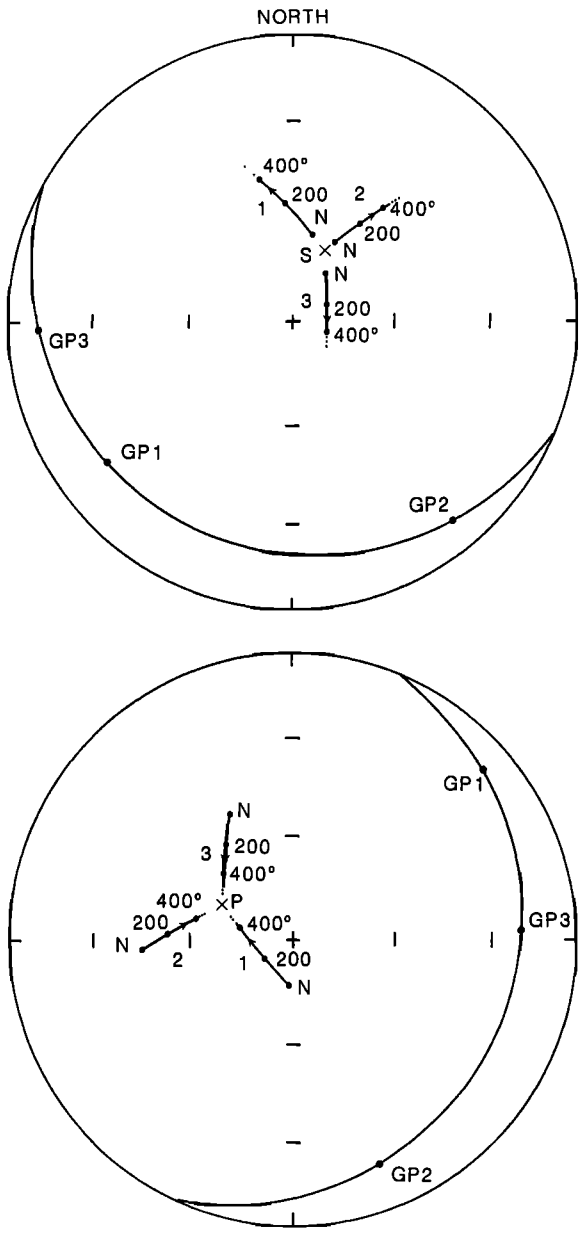


Fig. 3. Idealized example illustrating the use of demagnetization planes to solve for overlapping components of primary and secondary magnetization. In the upper equal-area net the secondary component (S) is found by solving for the "back" intersection of the three great circles defined by thermal demagnetization data (uncorrected for tilt) from three specimens. Data from each specimen lie on an arc (labeled by N (untreated natural remanent magnetization) and temperature steps) which in turn defines a plane and its pole (GP1, etc.) The intersection (S) is found by solving for the pole of the great circle defined by GP1, GP2, and GP3. In the lower diagram, application of the tilt corrections to the data shown above gives the primary magnetization direction (P). The "forward" intersection of the demagnetization planes defines the primary component.

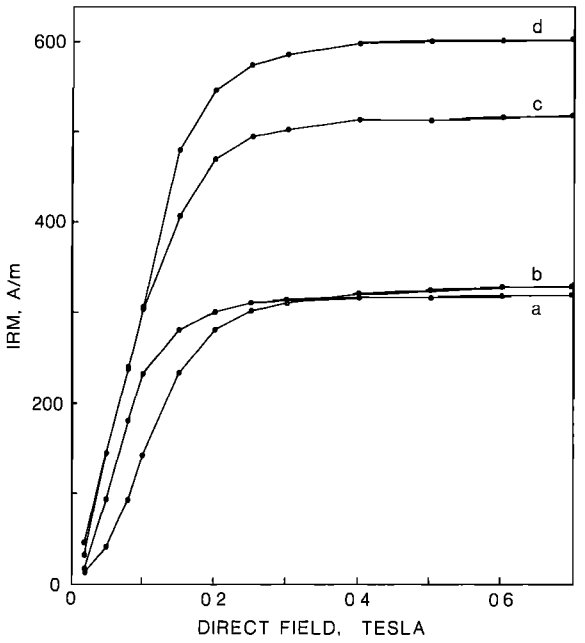


Fig. 4. Acquisition of isothermal remanent magnetization (IRM) in Lower Cretaceous (Neocomian and Barremian) volcanogenic rocks of the Koyukuk terrane: curve a, volcaniclastic rock; curve b, pillow basalt; curve c, lava flow; curve d, hypabyssal intrusive rock.

parameters for the *Bingham* [1974] distribution, we interpolated between the values given in Table 1 of *Mardia and Zemroch* [1977]. The oval of 95% confidence about the mean intersection was obtained from the Bingham precision parameters according to equations given by *Onstott* [1980] and *Bingham* [1974].

Neocomian-Barremian Volcanic Rocks

We examined thin sections of several representative rock types from the Lower Cretaceous volcanogenic sequence. The pillow basalt, tuffs, andesite flows, and volcaniclastic rocks have all retained their primary textures, but the rocks also show alteration by low-grade metamorphism. In some examples, such as the pillow basalt at site 5, the original plagioclase crystals have been replaced by epidote and calcite. Chlorite and epidote are pervasive throughout the volcanic sequence. The thin sections show no evidence of penetrative deformation, such as grain dislocations or stretching. The most abundant magnetic mineral is magnetite as interpreted from diagrams of the acquisition of isothermal remanent magnetization (IRM). The plateau in the IRM curves above 0.2. T indicates the magnetic saturation of magnetite (Figure 4). A slight rise in the IRM curves beyond 0.2 T is due to a high coercivity mineral, probably hematite in low concentration.

We have selected five examples from the various volcanogenic rock types in order to characterize the results of the demagnetization treatments. In general, specimens from the dacite flows, welded tuffs, and vitrophyre lost only moderate fractions of their natural remanent magnetizations (NRM) during af demagnetization (Figure 5*a*). Lesser resistance to af treatment was observed for the pillow basalts and volcaniclastic rocks. In most cases, however, the peak af of 100 mT was insufficient to reduce the NRM to below 10% of its initial value.

During thermal demagnetization the highest unblocking

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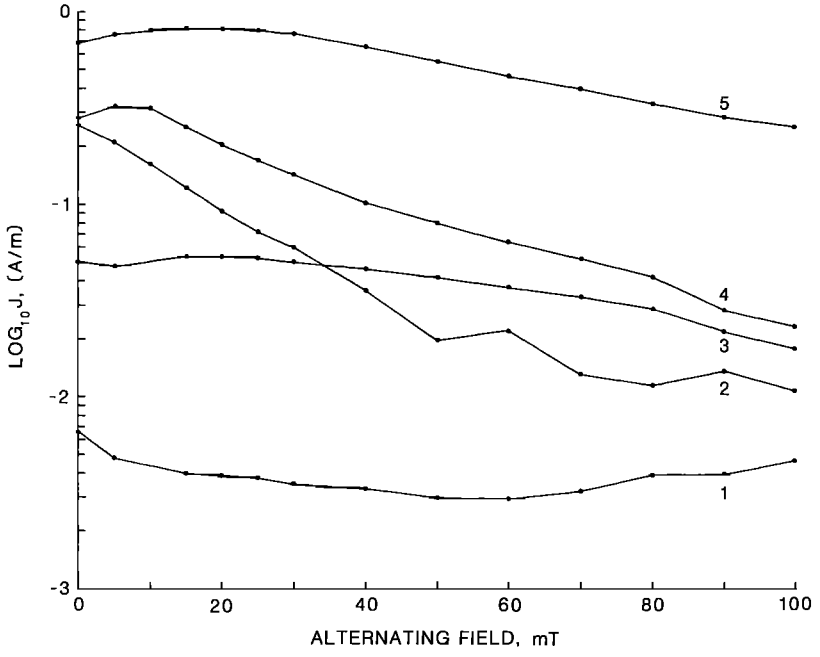


Fig. 5*a*

Fig. 5. Decay of natural remanent magnetization (J) during stepwise demagnetization of Koyukuk volcanogenic rocks: (*a*) alternating field demagnetization and (*b*) thermal demagnetization. Open circles denote onset of erratic directional changes. Curve 1, vitrophyre, location 3; curve 2, volcaniclastic rock, location 1c; curve 3, welded tuff, location 2a; curve 4, pillow basalt, location 6; curve 5, dacite flow, location 2b.

temperatures (650°–675°C) were exhibited by the vitrophyre, which contains hematite (Figure 5*b*). The welded tuffs and dacite flows usually exhibited a narrow range of unblocking temperatures between 550° and 610°C, where the majority of the magnetization was removed. The pillow basalts showed an unblocking temperature range between 550° and 585°C that was also narrow but was slightly lower than the range for the more silicic volcanic rocks. Remanent magnetizations of the volcaniclastic rocks gave a broad range of unblocking temperatures between 200° and 650°C.

From examination of the orthogonal vector diagrams, we found that the NRM usually consisted of three magnetic components. During the first two steps of the af and thermal demagnetization runs, the component that was removed usually had a nonsystematic direction of magnetization, perhaps caused by the coring and sawing procedure. The second component, typically removed by the 350° heating step, appears to be a well-defined overprint that masks the underlying component of magnetization. We could remove the overprint more effectively using thermal demagnetization, as af treatment usually failed to produce linear decay of the magnetization to the origin of the orthogonal vector diagram (Figure 6). Therefore all remaining specimens were thermally demagnetized, and these results were used in the final analysis of the magnetization. The same program of heating steps that was used on the pilot specimens was applied to the remaining specimens.

Our first step in studying the overprint direction was to fit least squares lines to the endpoints of the magnetic vectors obtained from the range of heating steps roughly between 80° and 300°. The dominant direction (not corrected for tilt of the bedding) is inclined steeply downward to the north. However, about 15% of the specimens exhibited an overprint of reversed polarity with steep inclinations in the upward, southern quadrant. Examination of the orthogonal vector diagrams suggested that there is some overlap between the unblocking temperature distributions of the overprint and the underlying magnetization. Such overlap introduces a large amount of directional scatter into the overall collection of lines corresponding to the overprint magnetization. To avoid this source of error, we fit least squares planes to the demagnetization results that were obtained from the full range of heating steps beyond

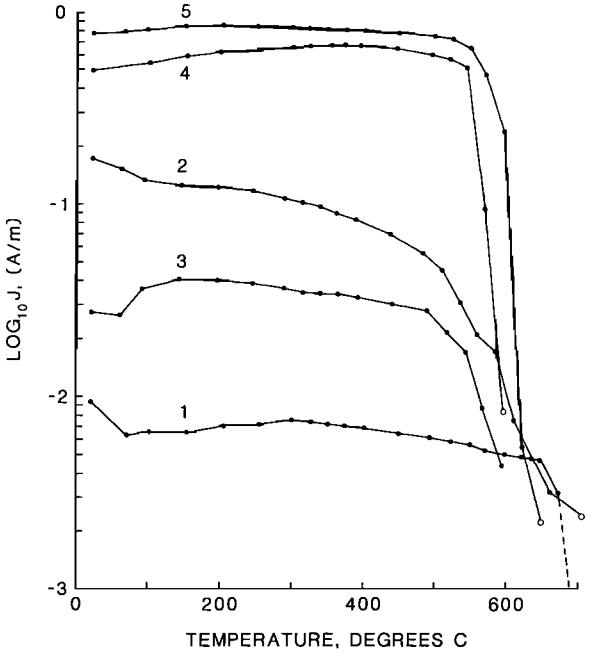


Fig. 5*b*

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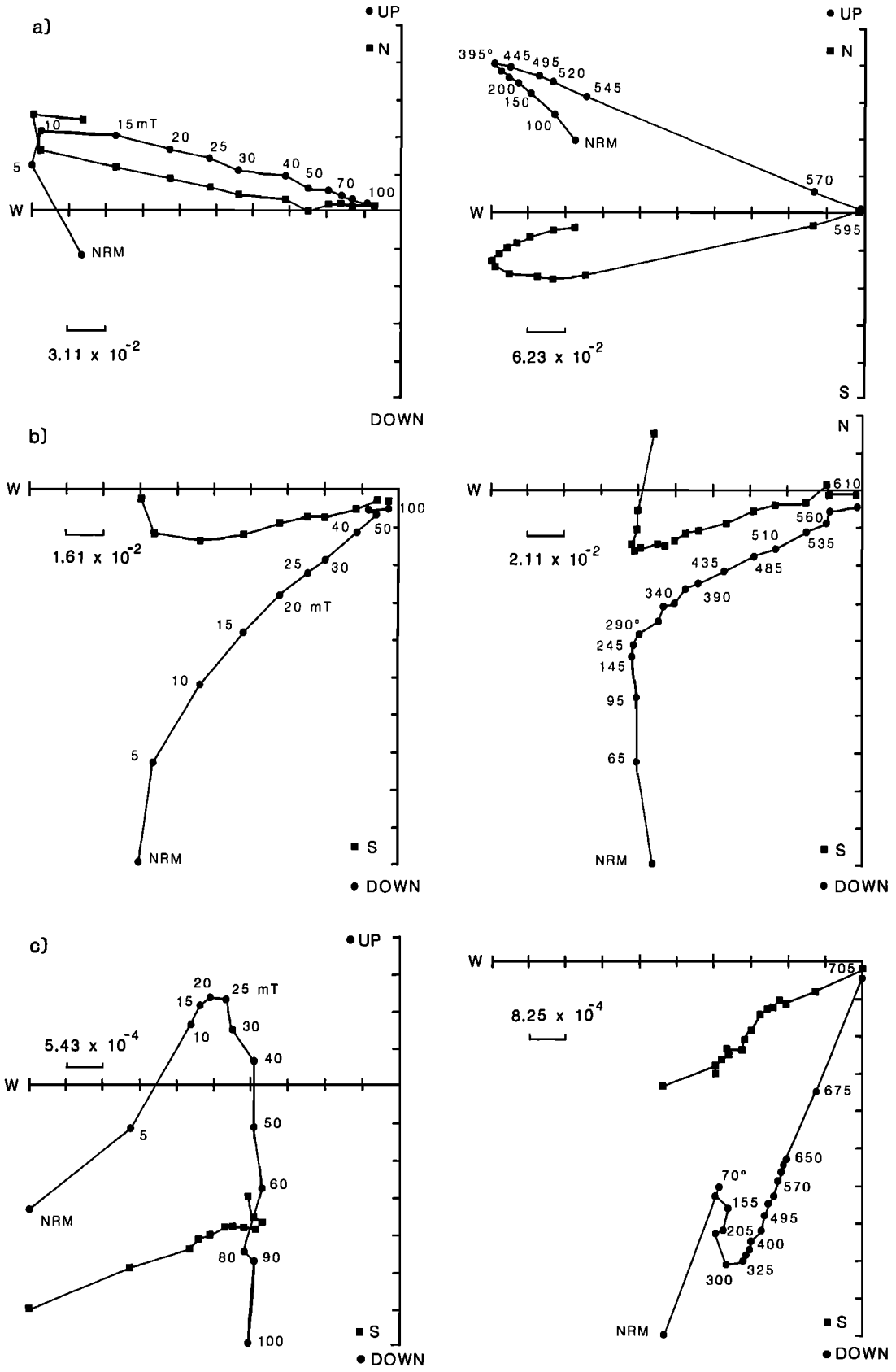


Fig. 6. Orthogonal vector diagrams showing results of alternating field demagnetization (left) and thermal demagnetization (right) of companion specimens from the Lower Cretaceous volcanogenic rocks: (*a*) pillow basalt, location 6, (*b*) volcaniclastic rock, location 1c, (*c*) vitrophyre, location 3, (*d*) dacite flow, location 2b. Circles are vector components of the magnetization (not corrected for tilt of the bedding) projected into the vertical plane; squares are projected into the horizontal plane. Bars denote scaling in amperes per meter.

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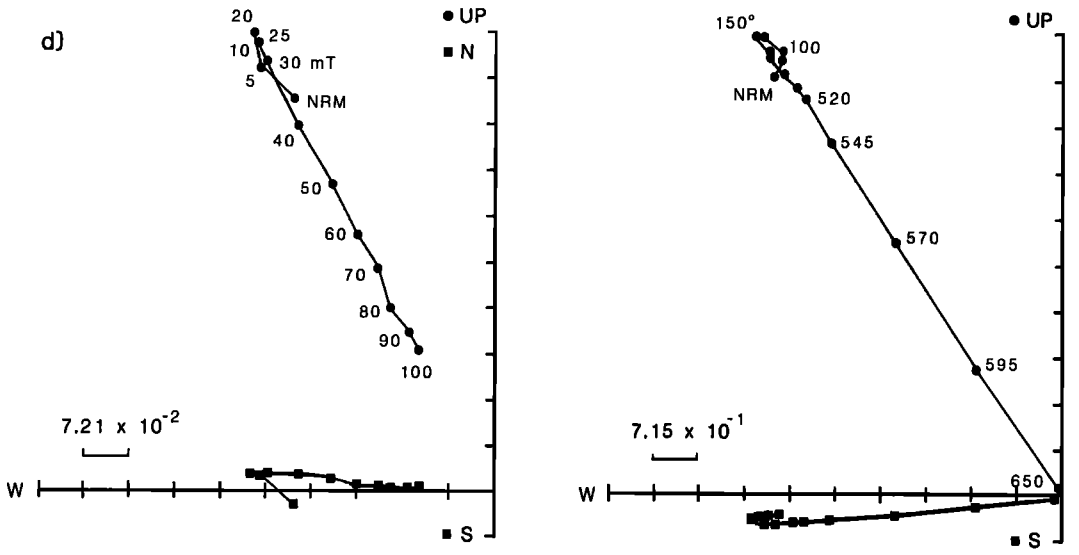


Fig. 6. (continued)

80°C (Figure 7). If the overprint was acquired after the regional deformation occurred, then the intersection of the demagnetization planes (not corrected for tilt of the bedding) should define the overprint quite accurately (Figure 3). This interpretation assumes that a prefolding primary magnetization exists and that the dispersion of the primary vectors from the various tilted localities is greater than the dispersion of the postfolding overprint. In Figure 8*a* the distribution of poles from the uncorrected demagnetization planes forms a girdle along a great circle, to which a least squares plane was fitted. The pole to the best fitting plane defines the intersection of the demagnetization planes and hence defines the mean direction of the overprint.

From the mean direction of the postfolding overprint, we can set limits on the age of the remagnetization episode. The 95% confidence oval of the overprint encloses direction corresponding to middle Tertiary paleomagnetic poles from the North American craton, possibly as old as the Eocene pole of *Diehl et al.* [1983], and it also encloses the present geocentric axial dipole field (Figure 8*b*). Since the overprint has both normal and reversed polarity, we can rule out Brunhes age (Brunhes Normal Polarity Chron: <0.7 Ma) viscous remanent magnetization. The overprint is therefore consistent with the geomagnetic dipole field throughout the middle Tertiary and early Pleistocene, assuming that the region has not been uniformly tilted since the major folding occurred. The high blocking temperatures associated with the overprint point to a regional thermal event of major importance, most likely coincident with the eruption of the Tertiary volcanic rocks. We can rule out the Indian Mountain pluton as the thermal source of the overprint because the age of the intrusion places it within the Cretaceous long normal polarity interval [*Harland et al.*, 1982]. The dual polarity and mean direction of the overprint are incompatible with a mid-Cretaceous thermal event.

As indicated by the examples (Figure 6), thermal demagnetization above 350° revealed a stable component of remanent magnetization, which in each case decayed along a well-defined line trending approximately toward zero intensity. Applying principal component analysis to the high-temperature steps, we determined characteristic magnetizations for all but 28 of the 189 specimens from the Neocomian-Barremian rocks. We typically used five to seven heating steps to define the least squares line corresponding to the characteristic magnetization, although as many as 15 steps or as few as three steps were used in a small percentage of the collection. We rejected results from specimens which gave very poorly defined lines, were completely overprinted, or had magnetizations that failed to decay linearly on the orthogonal vector diagrams.

After calculating the characteristic magnetizations from the acceptable specimens, we plotted the results from each cooling unit on an equal-area projection to examine the internal consistency of the magnetic directions. Unusually large directional

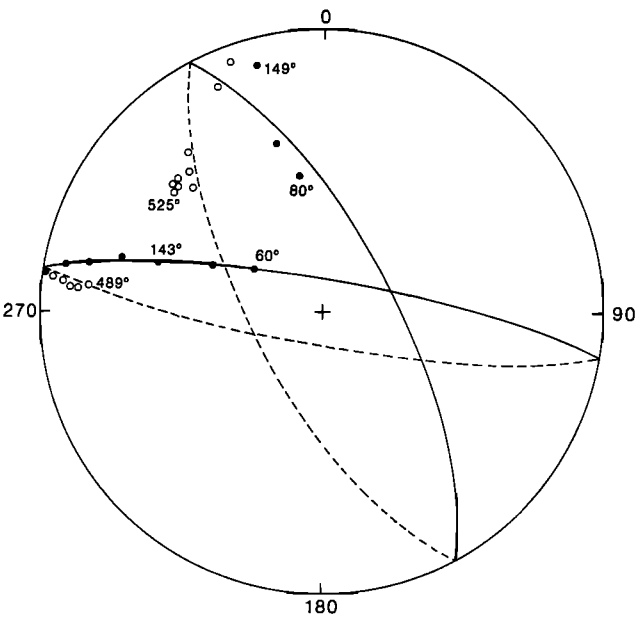


Fig. 7. Examples of demagnetization planes obtained by fitting a great circle to thermal demagnetization data from each of two specimens from site 6, units 1 and 2. Equal-area projection of magnetization directions after sequential heating steps; temperatures in degrees Celsius. Open circles and dashed lines lie in the upper hemisphere; solid circles and solid lines lie in lower hemisphere.

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scatter was observed in five of the 29 units, with the scattered results being mainly from sites 2b and 3. In most cases, the large scatter appeared to be due to smearing of the magnetization between the overprint and the high-temperature component, despite the apparent isolation of well-defined characteristic magnetizations. This observation indicates that the unblocking temperature distributions of the overprint and the underlying high-temperature component are inseparable at some localities. In a few other cases, one or two characteristic magnetizations would fall far from the main group of directions; we do not know whether these "flyers" are due to orientation errors or just represent natural extremes of the vector population. Due to poor internal consistency, two cooling units from site 3 were rejected altogether, and an additional 11 "flyers" were rejected. Overall, 147 specimens (77% of the collection) were retained for the analysis of the high-temperature component of magnetization.

For each cooling unit we calculated the mean magnetic direction of the high-temperature component as determined by least squares lines and assigned 95% confidence limits according to the statistical method of *Fisher* [1953] (Table 2). The unit mean directions are quite scattered when plotted before being corrected for tilt of the bedding (Figure 9*a*). The angular dispersion decreased significantly after the tilt corrections were applied, as the Fisher precision parameter increased by a factor of 3.3 (Figure 9*b* and Table 3). This improvement strongly indicates that the high-temperature component was acquired before the major regional deformation occurred. Also, the 10 reversed-polarity units are brought closer to antiparallelism with the normally polarized units after all were corrected for tilt of the bedding. However, the mean directions of the groups differ from antiparallelism by 21°, which would not be expected from a true sampling of the time-averaged geomagnetic field. Another weakness in the data distribution of Figure 9*b* is the apparent lack of circular symmetry and centered concentration, as would be expected from the near-Fisherian distribution of most overprint-free paleomagnetic data sets [*Fisher*, 1953]. Instead, the unit mean directions fall into groups that are defined by the different structural corrections.

These departures from a representative sample of the ideal time-averaged geomagnetic field might arise from (1) localized rotations that have introduced offsets in declination, and (2) incomplete removal of the overprinted magnetization. Sites 5 and 6, which include about a third of the cooling units, are in a structurally complex area, and localized rotations are to be expected (Figure 2). However, the structure is more coherent between sites 1 and 4, so we do not expect large variations in paleodeclination due to localized block rotations in the northern area. Instead, the severity of overprinting seen in the demagnetization experiments suggests that incomplete removal of the magnetic overprint is the major source of error.

Analysis of demagnetization planes under certain conditions, will successfully retrieve the primary magnetization from severely overprinted specimens [*Halls*, 1976; *Onstott*, 1980]. In a study such as ours, success of the method depends on having a reasonable variety of bedding attitudes and two components of magnetization: one acquired before folding, the other after folding. The method also assumes that apparent rotations due to incomplete corrections for folding about more than one horizontal axis are negligible. During the course of demagnetization of each specimen, the resultant magnetization vectors will lie within the plane defined by the overprint and the underlying primary component. If the variation of structural corrections is such that the application of tilt corrections causes the angular dispersion of the rotated overprint to be substantially greater than the angular dispersion of the restored primary component, then the tilt-corrected demagnetization planes will intersect at the primary magnetization vector (Figure 3).

Schmidt [1985] cautioned that solutions to remagnetization

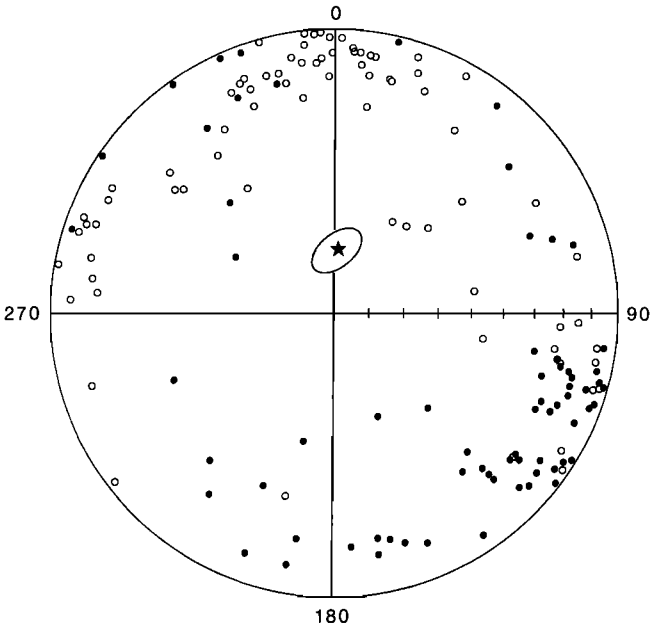


Fig. 8*a*

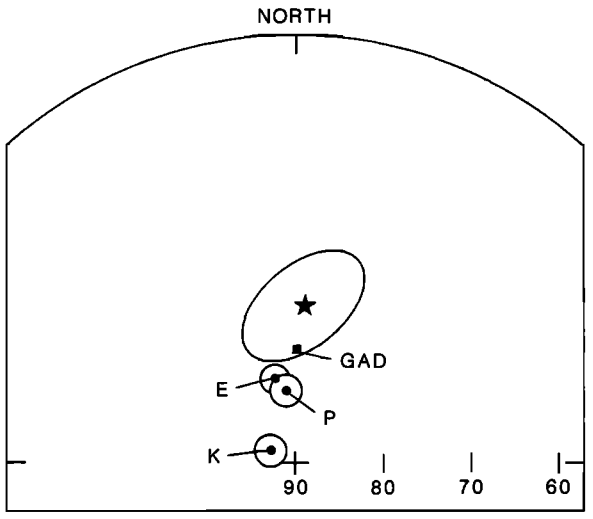


Fig. 8*b*

Fig. 8. (*a*) Mean magnetic direction (star) and 95% confidence limit of overprint magnetization in the Lower Cretaceous volcanogenic rocks, as derived from poles to demagnetization planes not corrected for tilt of the bedding (Table 4). Poles to demagnetization planes from 147 individually thermally demagnetized specimens are indicated by solid circles (lower hemisphere of equal-area projection) and open circles (upper hemisphere). (*b*) Comparison of mean overprint magnetic direction (star) with the present geocentric axial dipole direction (GAD) and magnetic directions corresponding to Eocene (E), Paleocene (P) [*Diehl et al.*, 1983], and Cretaceous (K) [*Gromme*, 1984] reference poles from stable North America. Ninety-five percent confidence limits are shown.

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TABLE 2. Paleomagnetic Results From Cretaceous Rocks of the Yukon-Koyukuk Province, Alaska

Site	Unit	Lithology	Treatment	N/N_c	k	α_{95}	I	D	I_c	D_c	λ	ϕ	A_{95}
1a	1	vc	300°–610°	8/8	330	3.1	67.7	290.1	68.9	66.6	55.3	285.1	4.8
1b	1	pb	350°–540°	5/5	14	20.9	−39.9	63.3	−53.9	251.9	38.4	295.7	25.7
1c	1	vc	330°–610°	5/5	226	5.1	25.1	253.7	68.0	50.5	60.8	301.1	7.9
1d	1	vc	300°–580°	4/4	71	10.9	29.9	291.7	70.3	322.7	68.6	101.0	17.9
	2	vc	330°-530°	3/3	774	4.4	65.9	285.0	67.9	84.1	47.4	273.1	6.9
	3	ms	320°–515°	4/4	1067	2.8	65.5	272.6	68.9	97.8	43.6	262.0	4.4
2a	1	wt	340°–545°	4/4	181	6.9	−5.3	268.9	−48.1	266.6	27.8	285.8	7.6
	2	wt	300°–580°	8/10	32	9.9	−18.0	281.2	−60.3	288.4	30.4	261.4	13.5
2b	1	ms	325°–570°	6/8	29	12.7	−18.9	263.7	−60.8	254.2	43.7	288.9	17.6
	2	wt	290°–515°	2/6	*	*	−31.7	278.8	−74.2	290.1	46.5	248.0	*
	3	lf	350°–550°	2/7	*	*	−36.3	265.6	−77.8	242.1	66.0	265.6	*
	4	lf	330°–625°	5/6	32	13.7	−37.8	271.6	−80.7	264.0	61.7	246.7	26.9
3	1	lf	330°–625°	0/8									
	2	lf	350°–550°	5/8	77	8.8	51.2	209.1	63.5	260.9	36.9	144.9	12.6
	3	lf	300°–550°	0/8									
	4	lf	350°–570°	6/8	30	12.4	64.1	222.5	61.5	292.7	47.2	117.1	17.3
	5	lf	350°–625°	3/7	34	21.6	61.3	265.7	43.1	304.0	36.4	94.5	22.3
	6	vt	250°–675°	6/6	21	14.9	71.2	230.6	59.7	308.1	51.7	100.4	20.2
4a	1	tf	300°–540°	4/4	674	3.5	66.6	297.9	71.6	34.4	71.2	310.1	5.9
4b	1	tf	150°–550°	7/7	67	7.4	61.9	316.8	62.2	22.4	64.3	346.1	10.4
5	1	pb	400°–625°	7/8	48	8.8	−30.4	260.0	−53.0	8.3	9.6	198.1	10.5
6	1	hi	200°–620°	7/7	170	4.6	30.8	21.2	81.1	209.5	49.8	192.0	8.9
	2	dk	200°–575°	8/8	33	9.8	48.0	25.9	63.8	197.4	22.1	192.1	14.3
	3	hi	250°–575°	8/8	118	5.1	36.1	9.3	71.9	238.8	40.3	167.3	8.6
	4	hi	250°–610°	7/8	27	11.7	35.1	12.0	74.0	235.2	42.4	171.6	20.9
	5	lf	300°–650°	7/8	30	11.2	26.3	12.1	79.6	266.3	57.9	164.8	21.8
	6	lf	450°–625°	5/5	9	26.7	−14.7	165.2	−54.0	117.7	42.4	106.1	33.2
	7	lf	310°–625°	3/3	32	22.1	33.9	31.4	75.8	171.1	39.1	210.4	40.7
	8	lf	400°–600°	6/8	23	14.2	−6.7	216.0	−69.9	244.9	57.0	284.4	23.5
7a	1	ms	10–70 mT	11/11	54	6.3	83.9	235.5	63.1	62.9	50.6	296.9	8.9
7b	1	ms	10–60 mT	17/17	63	4.5	72.6	256.5	59.2	48.9	52.4	314.9	6.0
7c	1	ms	10–80 mT	17/17	15.8	9.3	35.8	176.3	29.7	355.3	39.7	31.7	8.2

Treatment refers to temperature range of thermal demagnetization in degrees Celsius, or range of applied alternating field in millitesla; N_c, number of specimens collected; N, number of specimens averaged; k, *Fisher* [1953] precision parameter; α_{95}, 95% confidence limit for directions in degrees; I and D, inclination and declination without correction for tilt of the bedding, in degrees; I_c and D_c, inclination and declination of tilt-corrected directions, in degrees; λ and ϕ, latitude and east longitude of virtual geomagnetic pole (tilt-corrected), in degrees; A_{95} 95% confidence limit for virtual geomagnetic pole. Lithology symbols: dk, dike; hi, hypabyssal intrusion; lf, lava flow; ms, mudstone and sandstone; pb, pillow basalt; tf, waterlain tuff; vc, volcaniclastic rock; vt, vitrophyre; wt, welded tuff.

*Too few samples to calculate statistic.

problems involving two components of magnetization can be biased if the dispersions of the underlying vector populations are not significantly different. In the case of our study, application of the tilt corrections to the overprint vector yields a Fisher precision parameter k of 2, substantially smaller than the $k = 20$ expected from normal secular variation of the primary component. Moreover, the tilts are roughly evenly distributed between east and west. Therefore we do not expect any substantial bias in solving for the intersection of demagnetization planes.

In our analysis the same demagnetization planes that were used in locating the overprint direction were corrected for tilt; then for each cooling unit, the poles to the demagnetization planes of the individual specimens were averaged together, yielding 27 mean poles. The resultant distribution of the mean of poles to planes falls near a great circle, the pole of which in turn defines the primary component of magnetization (Figure 10*a* and Table 4). Because the majority of demagnetization planes cross through the northeast and southwest quadrants of the lower hemisphere, the long axis of the 95% confidence oval is aligned to the northeast. The long axis of the oval is quite large partly due to the uneven distribution of poles within the girdle defined by the averaged demagnetization planes and partly due to the scatter about the great circle that best describes the girdle. Within some of the cooling units, the orientations of the specimen demagnetization planes are quite variable so that the confidence limits are poor for the resultant unit-mean poles to planes. This variability within cooling units indicates that the total magnetization is more complicated than a two-component system, as we must assume in the planes analysis. Certainly, there are variations in the postfolding overprint, as indicated by its dual polarity. Because the number of cooling units is only 27, the unit-mean demagnetization planes that are poorly defined increase the confidence limit of the overall unit mean more than one would prefer.

As an alternative, we have also calculated an overall mean direction for the primary magnetization by solving for the intersection of all demagnetization planes from the individual specimens ($N = 147$). Compared to the previous analysis, this calculation has the advantage of compensating better for the within-unit variability of the postfolding overprint. Moreover, the specimen mean reduces the weighting for the least stable units, represented by fewer specimens. A disadvantage is that the number of specimens is greater than the actual number of independent time samples of the primary component. As would be expected, the specimen-mean direction is very close to the unit mean, and the 95% confidence oval is smaller due to the greater N (Figure 10*b* and Table 4). Because the statistics calculated from the specimen means are based on an overestimate of the true number of independent time samples of

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the primary component, we consider the size of the corresponding confidence oval to be a minimum estimate. On the other hand, the average planes were poorly determined at some sites owing to large variability of the secondary components. Hence the site-mean confidence oval is considered the maximum estimate.

One of the purposes of the demagnetization plane analysis was to reduce the 21° departure from antiparallelism exhibited by the normal and reversed polarity units after least squares lines were fit to the demagnetization data (Table 3). By performing the planes analysis separately for each polarity group (Table 4 and Figure 10*b*), we demonstrate that the normal and reversed means move markedly closer into antiparallelism (8° departure) as compared to the outcome of the lines analysis. This result, a positive "consistency of reversals test," justifies our use of demagnetization planes and indicates the presence of residual secondary magnetizations despite detailed thermal treatments.

The primary magnetization in the Koyukuk volcanogenic rocks transforms to a paleomagnetic pole in the Sea of Okhotsk (Figure 11). For purposes of comparison, we also show two reference poles from the North American craton of late Jurassic and Cretaceous age. The older reference pole (64.6°N, 164.2°E, $\alpha_{95} = 3.9°$) is from the upper part of the Morrison Formation of late Jurassic (Tithonian) age, where the original pole determined by *Steiner and Helsley* [1975] has been corrected for clockwise rotation (3.8°) of the Colorado Plateau [*May and Butler*, 1986; *Bryan and Gordon*, 1986]. The younger reference pole (67.3°N, 186.3°E, $\alpha_{95} = 3.4°$) is from a compilation of Cretaceous results from rocks with isotopic ages between 87 and 135 Ma [*Gromme*, 1984]. During this 48-m.y. period, which overlaps the ages spanned by the Neocomian-Barremian volcanic rocks, North America was essentially stationary relative to the paleomagnetic axis. We consider the Cretaceous pole to be the best reference pole currently available for our study of the Koyukuk region. We also show in Figure 11 the paleomagnetic pole (49.1°N, 213.9°E, $\alpha_{95} = 5°$) obtained by *Halgedahl and Jarrard* [1987] from the Kuparuk River Formation of Early Cretaceous (Neocomian) age on the North Slope.

From the Cretaceous reference pole the expected paleolatitude of our sampling locality in the Koyukuk basin is 82° ± 3°N. As determined from our paleomagnetic results, the mean paleolatitude of the study area is 69°N, and the range of paleolatitudes encompassed by the 95% confidence oval is 60°–75°N. This confidence limit corresponds to the specimen mean and is thus a more optimistic estimate of the uncertainty in the mean as compared to the limit we calculated from the site mean. We emphasize that the mean pole position from our study is statistically very stable, whether the specimen mean or the site mean is used. Note that the range of paleolatitudes corresponds to the extremes of the elongate confidence oval (closest and farthest points on the confidence oval as measured from the sample site) and does not represent a 95% confidence limit in the strict sense defined by *Fisher* [1953]. As can be inferred from Figure 11, a circle centered on the sample site and drawn through the Koyukuk paleomagnetic pole will envelop less than 50% of the area inside the confidence oval, indicating that the lower end of the paleolatitude range should be favored over the higher end. Hence we give this paleolatitude range as merely a rough estimate of the 95% confidence limits. From the confidence limits of the paleomagnetic data, we infer that the Lower Cretaceous volcanic rocks of the Koyukuk basin formed in the high Arctic region. The mean paleolatitude suggests that the Koyukuk terrane has shifted to a higher latitude relative to the continental interior of North America.

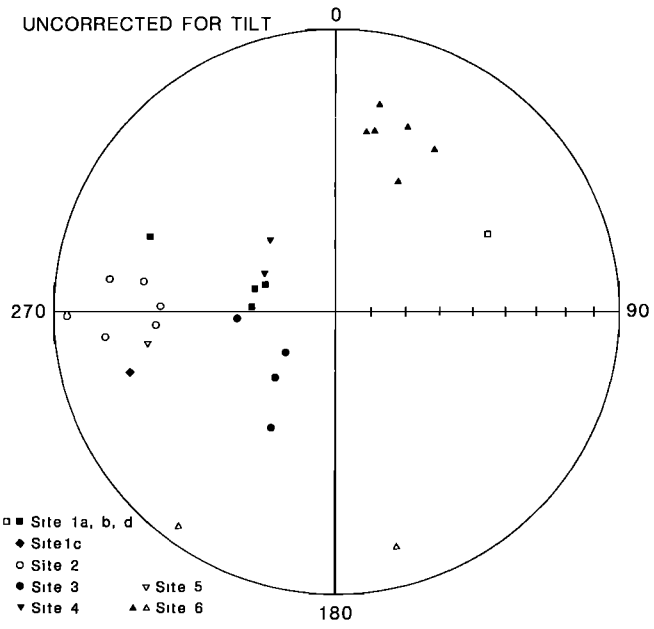


Fig. 9*a*

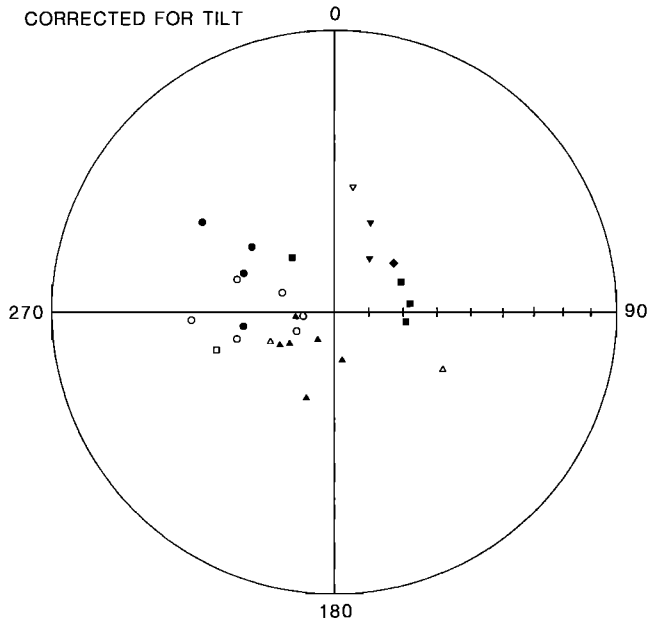


Fig. 9*b*

Fig. 9. Mean magnetic directions from individual cooling units and sections from the Lower Cretaceous volcanogenic rocks (Table 2). (*a*) Not corrected for tilt of the bedding; (*b*) corrected for tilt. Magnetic directions were calculated by fitting least squares lines to the high-temperature steps of the thermal demagnetization data. Solid symbols are in the lower hemisphere of the equal-area projection; open symbols are in the upper hemisphere.

Albian Sedimentary Rocks

We sampled three short stratigraphic sections in the Albian turbidites along the east bank of the Koyukuk River near

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TABLE 3. Analysis of Thermal Demagnetization Lines From the Lower Cretaceous Volcanic Rocks of the Koyukuk Terrane, Alaska

	N	k	α_{95}	I	D	λ	ϕ	A_{95}
Sites 1–6, High-temperature component								
All units, not corrected for tilt of the bedding	27	3	21.0	69.1	23.5	82.6	341.6	29.9
All units, corrected for tilt of the bedding	27	10	9.3	86.1	49.1	69.4	223.0	16.3
Normal polarity units, corrected for tilt	17	12	10.8	84.1	310.2	70.5	182.4	19.4
Reversed polarity units, corrected for tilt	10	10	16.0	−74.2	268.9	57.3	260.4	25.3

N, number of units averaged; k, *Fisher* [1953] precision parameter for magnetic direction; α_{95}, 95% confidence circle, in degrees; I and D, inclination and declination of remanent magnetization direction, in degrees; λ and ϕ, latitude and east longitude of north paleomagnetic pole, in degrees; A_{95}, 95% confidence limit of paleomagnetic pole.

Hughes (Figure 2). Overall, the sections span a horizontal distance of approximately 0.5 km. Sites 7a–7c are in a tight fold where the north dipping beds pass through the vertical plane and become overturned by about 25° (Table 1). The turbidites are finely laminated so that individual specimens usually encompass several layers of gray silt and fine sand. The intensities of NRM were typically in the range from 10^{-3} to 10^{-2} A/m.

The af demagnetization at fields above 20 mT usually reduced the magnetization to one half of the initial NRM, and fields above 60 mT produced erratic changes of the magnetic direction (Figure 12*a*). Thermal demagnetization of adjacent specimens yielded smooth demagnetization paths up to the 450°C step where erratic directional changes became apparent; above 500°C the intensity of magnetization usually increased sharply (Figure 12*b*). Although both demagnetization methods produced similar directional changes at the lower steps in the runs, the thermal treatments apparently altered the magnetic minerals after the heating had progressed beyond 450°C. Therefore we selected af demagnetization as the better method to treat the remaining specimens from the turbidite beds. As indicated by orthogonal vector diagrams, af treatment up to 15 mT removed a soft component of magnetization, then beyond 15 mT, the treatments produced linear decay of the magnetization vector.

After fitting least squares lines to the high-coercivity part of each demagnetization path, we combined the results to obtain the mean magnetic direction of each stratigraphic section (Figure 13 and Table 2). Before being corrected for tilt, the mean direction from site 7c is unlike the steep and downward means from sites 7a and 7b. Correcting for tilt of the bedding does not significantly decrease the directional dispersion of the means, and we are unable to obtain a useful overall mean direction of magnetization from the Albian turbidites by using least squares lines. We note that at roughly 50% unfolding, i.e., correcting for half of the tilt, the dispersion of the three-site means is minimized (S. Halgedahl and R. Jarrard, written communication, 1988). A possible interpretation is that the magnetization was acquired during Late Cretaceous folding. If so, we cannot be sure of the true paleohorizontal at the time of magnetization lock-in, and the result does not reliably fix the paleolatitude of the Koyukuk basin.

Because the analysis of demagnetization lines had failed, we turned to the method of analyzing remagnetization circles by using least squares planes. The magnetization vectors of site 7c are streaked along the north-south meridian, probably due to incomplete removal of an overprint magnetization. As with the volcanic rocks, we reanalyzed the demagnetization data by fitting planes to the vectors from each af demagnetization run in an effort to compensate for the unremoved portion of the overprint. The demagnetization planes, however, failed to intersect with reasonable consistency. Either the overprint varies considerably over the region sampled or one of the sites has been tilted after overprinting occurred. Whatever the cause, we cannot determine a useful mean direction from the turbidites using planes analysis. We conclude that the Albian marine sedimentary rocks near Hughes no longer preserve the primary depositional magnetization and therefore cannot provide a reliable paleolatitude determination.

PALINSPASTIC RECONSTRUCTION OF NORTHERN ALASKA

Many current models for the tectonic history of Alaska and the northeast USSR are predicated on the concept of accreted microplates. Comparison of stratigraphic columns in the Brooks Range, Chukotka (USSR), and Wrangel Island (USSR) indicates these areas are part of a continental passive margin that is floored by Proterozoic crust [*Churkin and Trexler*, 1980; *Churkin et al.*, 1985]. The northern boundary of this block, which we designate as the "Arctic Alaska microcontinent," is the continental shelf in the Beaufort and Chukchi seas (Figure 1). The southern boundary of the Arctic Alaska microcontinent is marked by two belts of highly disrupted rocks of oceanic affinity. In Alaska the oceanic belt is the Angayucham terrane, which is composed of Devonian limestone and basalt, Mississippian chert, Permian tuff and limestone, and Upper Triassic and Lower Jurassic pillow basalt [*Jones et al.*, 1987]. This terrane lies south of the thrust sheets of the Brooks Range and is bounded on the south by the Yukon-Koyukuk province. Similar belts are found in the northeast USSR, where the South Anyuy and Oloy belts lie south of the Chukotka continental rocks. The Anyuy belt has been described as an Early Cretaceous suture between Chukotka and the Omolon continental block [*Fujita*, 1978; *Sweeney et al.*, 1978; *Seslavinskiy*, 1979; *Parfenov and Natal'in*, 1986]. *Box* [1985] views the Oloy zone [*Til'man et al.*, 1977] as a volcanic arc similar to the Koyukuk terrane.

The key element in Arctic reassemblies is the origin of the Canada Basin, which forms the abyssal plain between Alaska, the East Siberian Shelf, and the Canadian Arctic islands. Many believe that the Canada Basin formed by seafloor

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spreading and that continental microplates of Alaska and the northeast USSR moved outward from the basin as it formed. One class of models, first proposed by *Carey* [1955] and later modified by *Grantz* [1966], *Tailleur* [1969], *Hamilton* [1970] and others, proposes counterclockwise rotation of Arctic Alaska from an original position against the Sverdrup margin in Canada. These models require passive margins along the Beaufort and Sverdrup shelves and a transform fault in the East Siberian Sea. Others propose that the Alaskan margin has been a transform fault, essentially leaving Alaska in place as microcontinental blocks moved into the northeast USSR during seafloor spreading in the Canada Basin [*Herron et al.*, 1974; *Fujita*, 1978]. In a third group of models the Canada Basin is viewed as old oceanic crust, now entrapped by microplates carried in from the Pacific basin [*Churkin and Trexler*, 1980]. The ocean entrapment model implies that the Canada Basin is rimmed by very old continental margins, much older than the margins required by the rotation and transform models.

Geophysical evidence indicates a Cretaceous age for the seafloor of the Canada Basin, which favors the rifting models over the ocean entrapment model. Interpretation of seismic profiles across the shelves in Alaska and Canada suggests that the sedimentary apron is at least mid-Cretaceous in age [*Sweeney et al.*, 1978; *Grantz et al.*, 1981]. The substrate beneath the thick sedimentary cover of the Canada Basin has not been dated directly, although the depth profile and heat flow are consistent with seafloor being 90–125 m.y. old [*Lawver and Baggeroer*, 1983]. Magnetic anomalies are very weak in the Canada Basin, so the age of the oceanic crust and even the direction of spreading have not been determined with certainty [*Taylor et al.*, 1981; *Vogt et al.*, 1982].

Until recently, paleomagnetic studies of the Brooks Range were inconclusive regarding the pre-Mesozoic paleoposition of Arctic Alaska. Attempts to determine paleopoles from the Paleozoic Endicott and Lisburne Groups were unsuccessful due to pervasive overprinting of original depositional remanent magnetization in the Brooks Range [*Hillhouse and Gromme*, 1983]. In the same article we reported results from an oriented core taken from a deep borehole on the North Slope. The results appeared to be free of the overprint and were consistent with the counterclockwise rotation hypothesis. However, uncertainties in the age of magnetization and orientation of the core weakened the credibility of this result. Strong evidence in support of the rotational hypothesis has since been obtained by *Halgedahl and Jarrard* [1987] in their study of the Lower Cretaceous (Neocomian) Kuparuk River Formation on the North Slope. Their results are from oriented cores taken in two exploratory boreholes near Prudhoe Bay. The paleopole from the Kuparuk River Formation is most consistent with a tectonic reassembly that places the Arctic Alaska margin against the Sverdrup margin in pre-Neocomian time. According to the favored reconstruction the Arctic Alaska microcontinent rotated approximately 70° counterclockwise about a pivot point near the MacKenzie River delta, as proposed by *Grantz et al.* [1979]. Opening of the Canada Basin in Neocomian to mid-Cretaceous time is consistent with what is known about the provenance of Cretaceous sedimentary sequences in the Sverdrup basin and on the North Slope.

Today, the Yukon-Koyukuk province is in close structural contact with the Arctic Alaska microcontinent, the two provinces being separated from each other by the narrow Angayucham terrane. Ophiolites in the Angayucham terrane are possibly the remnants of an ocean basin that once separated the Koyukuk arc from the Arctic Alaska microcontinent. In the northeast USSR, ophiolites of the Anyuy belt may also be remnants of a Paleozoic and Mesozoic ocean that was closed

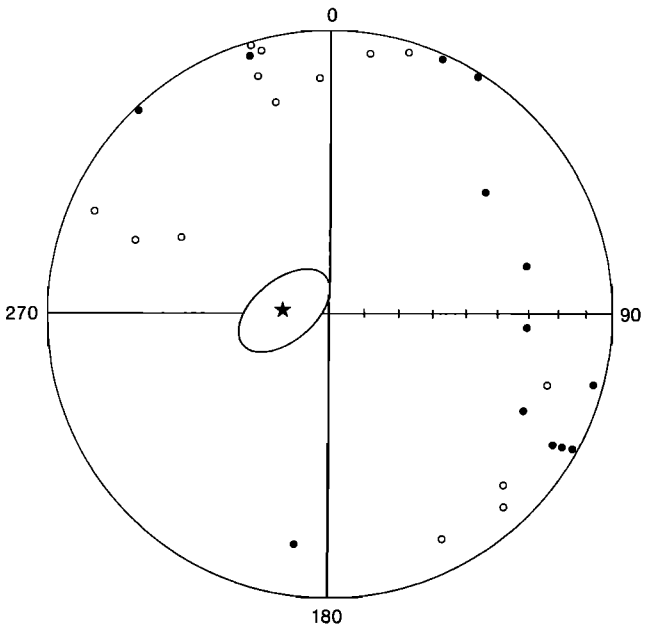


Fig. 10*a*

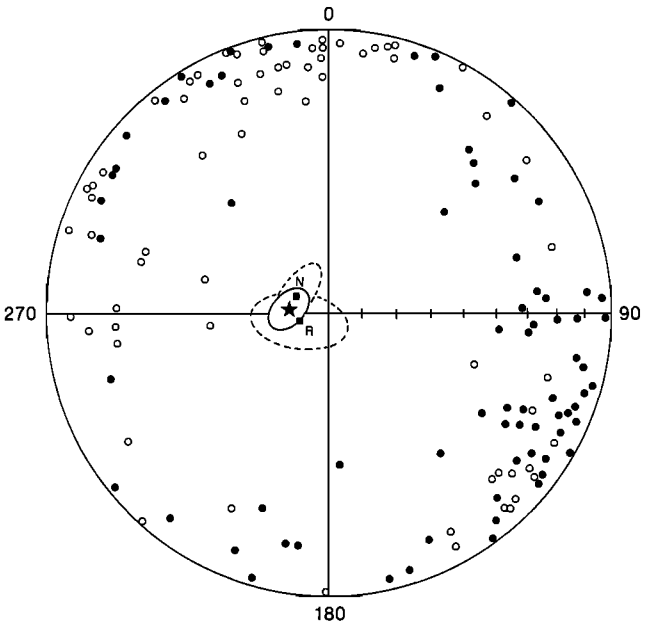


Fig. 10*b*

Fig. 10. (*a*) Mean direction (star) and 95% confidence limit of primary magnetization of the Lower Cretaceous volcanogenic rocks (corrected for tilt of the bedding; Table 4). The mean magnetic direction was derived from the mean poles to the thermal demagnetization planes obtained from the 27 cooling units. (*b*) Same analysis as in Figure 10*a* except that the mean direction was obtained from the 147 individual specimen demagnetization planes. Also shown are means derived from normal polarity specimens (N) and reversed specimens (R), grouped separately. Poles to demagnetization planes are denoted by solid circles (lower hemisphere) and open circles (upper hemisphere).

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TABLE 4. Analysis of Thermal demagnetization Planes From the Lower Cretaceous Volcanic Rocks of the Koyukuk Terrane, Alaska

	N	k_1	k_2	α_{min}	α_{max}	Oval Azimuth	I	D	λ	ϕ
Overprint magnetization (all specimens from sites 1–6, not corrected for tilt of the bedding)	147	−6.1	−1.4	4.7	7.9	49.0	72.1	3.3	81.1	13.8
High-temperature component (all units from sites 1–6, corrected for tilt of the bedding)	27	−8.1	−1.6	8.9	15.6	42.9	77.0	274.0	57.3	154.9
High-temperature component (all specimens from sites 1–6, corrected for tilt of the bedding)	147	−6.8	−1.3	4.4	7.0	49.7	79.1	276.2	60.2	159.6
Normal polarity specimens (sites 1–6, corrected)	94	−8.3	−2.6	4.4	10.7	37.6	79.5	298.7	67.6	152.2
Reversed polarity specimens (sites 1–6, corrected)	53	−5.6	−1.5	8.2	14.8	97.1	−81.2	73.8	56.9	174.0

N, number of specimens or units averaged; k_1 and k_2, *Bingham*'s [1974] concentration parameters; α_{min} and α_{max}, minor and major axes of 95% confidence oval, in degrees; oval azimuth, azimuth of major axis of confidence oval, in degrees; I and D, inclination and declination of best intersection of demagnetization circles, in degrees; λ and ϕ, latitude and east longitude of north paleomagnetic pole corresponding to I and D, in degrees.

by the collision of the Chukotka and Omolon continental blocks. Hence the Angayucham and Anyuy terranes are perhaps part of a major suture of Cretaceous age [*Box*, 1985], which marks the collision of the Arctic Alaska microcontinent with the interior terranes of Alaska and Siberia.

As a starting point for reconstructing the Neocomian position of the Koyukuk terrane, we assume the rotational hypothesis to be true for the movement of Arctic Alaska, following the conclusions of *Halgedahl and Jarrard* [1987]. This places northern Alaska and Chukotka against the Sverdrup margin in Canada during the Early Cretaceous (Neocomian) as shown in Figure 14. We consider several possible paleopositions of the Koyukuk terrane, given the Cretaceous reference pole for North America and the paleolatitude range (60°–75°N) obtained from our study (see the lettered options in Figure 14). The first option (A) is to have the Koyukuk terrane separated from Arctic Alaska by a narrow seaway, essentially leaving the terrane in its present position relative to the North American craton. This reconstruction, however, places the Koyukuk terrane at a considerably higher latitude than the measured paleolatitude range. An original position against Chukotka (B) is consistent with the paleomagnetic data but requires strike-slip translation of the Koyukuk terrane prior to rotation of the Arctic Alaska microplate.

Other options include reconstructions that place the Koyukuk terrane in the northern Pacific basin on the side of the paleomagnetic axis opposite to the pre-drift position of Arctic Alaska (C, for example). Depending on where the Koyukuk terrane is placed along the 60°–75° paleolatitude band between the North American and Asian continental margins, this reconstruction requires a wide ocean (1000–3000 km) between the Koyukuk volcanic arc and the Arctic Alaska microcontinent during Neocomian time. Rapid consumption of the intervening ocean would be aided by opening in the Canada Basin in addition to northward convergence of oceanic plates from the Pacific basin [*Engebretson et al.*, 1985]. As another alternative (D), the original position of the Koyukuk terrane might have been along the North American margin, with the northward transport occurring along a right-lateral fault at the continental margin.

By Albian time (113–97 Ma), volcanism had ceased in the Koyukuk arc as orogenic detritus accumulated within the marine Koyukuk basin. The shutoff of volcanism and the general continuity of mid-Cretaceous basinal deposits across northern Alaska and northeastern Asia have been interpreted as marking the end of the subduction of oceanic crust between the Arctic Alaska microcontinent and the Yukon-Koyukuk province [*Box*, 1985]. However, paleolatitude determinations from Albian sediments in the Koyukuk basin are discordant with respect to the North American reference pole and indicate that the final tectonic assembly of the region may have occurred after 90 Ma [*Harris et al.*, 1987]. Although the Albian turbidites near Hughes are severely overprinted, the equivalent deposits in the Nulato quadrangle southwest of our

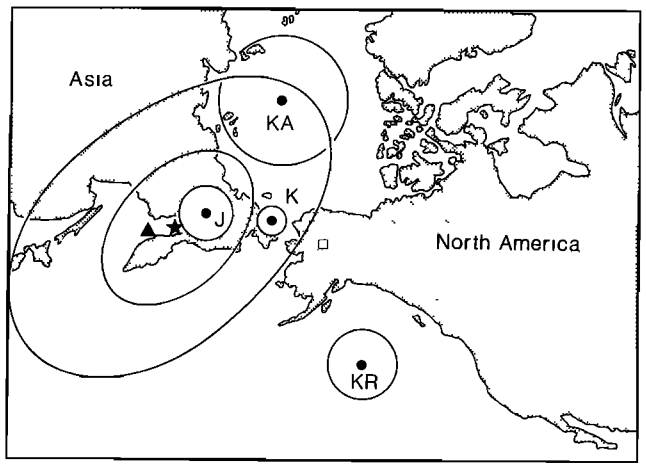


Fig 11. Comparison of paleomagnetic poles from Neocomian rocks of Arctic Alaska with reference poles from the North American craton. The North American reference poles (with 95% confidence limits) are from the Upper Jurassic (Tithonian) Morrison Formation (J) [*Steiner and Helsley*, 1975; *May and Butler*, 1986] and the Cretaceous (K) compilation of *Gromme* [1984]. The mean paleomagnetic pole from the Koyukuk terrane (square), which was derived from planes analysis of the specimen demagnetization planes, is indicated by a star, and the smaller oval is the 95% confidence limit (Table 4 and Figure 10*b*). The mean paleomagnetic pole and 95% confidence limit obtained from the analysis of cooling units are denoted by the triangle and the larger oval (Table 4 and Figure 10*a*). KR, paleomagnetic pole from the Kuparuk River Formation on the North Slope [*Halgedahl and Jarrard*, 1987]. KA, paleomagnetic pole from Albian sedimentary rocks of the Yukon-Koyukuk province (MLM from Table 2 of *Harris et al.* [1987]).

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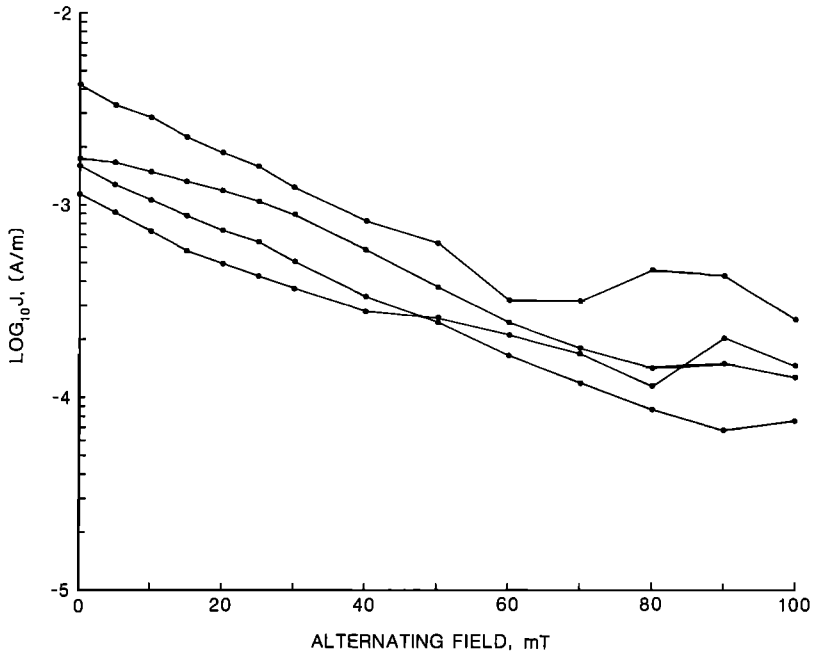


Fig. 12*a*

Fig. 12. Decay of natural remanent magnetization during demagnetization of representative specimens from the Albian turbidites. (*a*) Alternating field demagnetization; (*b*) thermal demagnetization; onset of erratic directional changes indicated by open circles.

study area satisfy reliability criteria which support a primary detrital origin for the magnetization. From these sedimentary rocks, *Harris et al.* [1987] determined a paleolatitude anomaly of 15° and inferred that the Koyukuk basin moved from the Pacific basin into higher latitudes (with respect to cratonic North America) during the interval 90–56 Ma. The upper age limit on the translation is constrained by paleomagnetic results of *Harris et al.* [1987] from Eocene volcanic rocks in the Koyukuk basin.

Because distinctive plutonic rocks and overlapping sedimentary rocks tie the Koyukuk terrane to the Arctic Alaska microcontinent by the Albian, the large amount of post-Albian northward drift proposed by *Harris et al.* [1987] is at odds with a simple rift origin for the Canada Basin. Large-scale translation of northern Alaska from the Pacific side after Albian time would have to be accommodated by tremendous shortening within the Koyukuk basin or within a yet undiscovered compressional zone in the Arctic Ocean. Although the structure of the Koyukuk basin is consistent with the occurrence of some compression, crustal shortening greater than 1000 km is very unlikely.

Given the paleomagnetic evidence that all of the Cretaceous terranes of northern Alaska most likely originated above 65°N, we cannot argue strongly for displacements of 15° or less, simply because the resolution of paleomagnetic data is poor at high latitudes. Low resolution of the paleomagnetic method above 65°N is due to the nature of the dipole field, whereby small errors in inclination translate into much larger errors (up to a factor of 2) in paleolatitude. Although confidence limits are useful for assessing the internal consistency of a given paleomagnetic result, the formal confidence limits do not include hidden sources of systematic errors. In sedimentary rocks deposited at 70°N, for example, a 5° systematic flattening of the inclination causes an apparent paleolatitude anomaly of 9°. Overprinting, bedding corrections, and inclination errors due to depositional or diagenetic processes can easily introduce errors in paleolatitude of the order of 10°.

CONCLUSIONS

Severe magnetic overprinting poses a serious problem in determining the Early Cretaceous paleolatitude of the Koyukuk terrane. We found that thermal demagnetization was more effective than the af method in stripping away the overprint from the Koyukuk volcanic rocks of Neocomian age. Even after thermal demagnetization, the volcanic rocks retained a portion of the overprint, necessitating the use of least squares planes analysis [*Halls*, 1976; *Kirschvink*, 1980] to

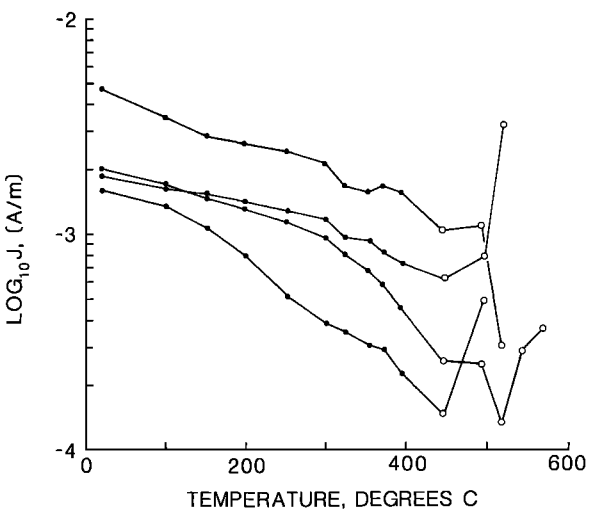


Fig. 12*b*

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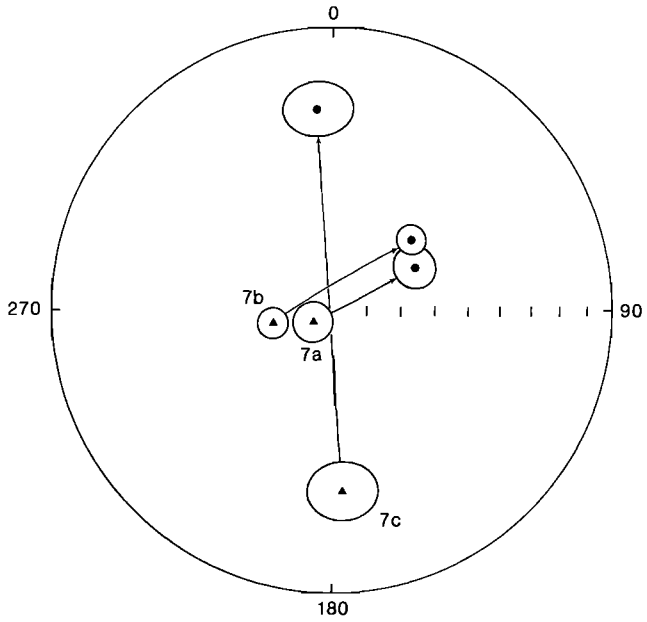


Fig. 13. Mean magnetic directions and 95% confidence limits from three af-cleaned sections in the Albian turbidites before (triangles) and after (solid circles) application of tilt corrections (Table 2). Equal-area projection, lower hemisphere.

obtain a more reliable estimate of the paleolatitude. By fitting planes to the demagnetization data and solving for the best intersection of the corresponding remagnetization circles, we obtained a mean paleopole at 60.2°N, 159.6°E. The mean paleolatitude is 69°N with the 95% confidence ellipse ranging from 60° to 75°N. This analysis was successful because the tilt corrections of the sampled strata varied sufficiently to ensure a strong intersection of the remagnetization circles.

As determined from Cretaceous paleopoles from stable North America, the predicted paleolatitude of the Koyukuk basin was 82°N, so we infer that the Koyukuk volcanic arc has moved poleward relative to the craton by approximately 1500 ± 800 km. *Harris et al.* [1987] determined a similar paleolatitude anomaly (15°) from Albian turbidites southwest of our study area, suggesting that poleward drift of the Yukon-Koyukuk province occurred after 97 Ma. However, post-Albian movement of the Koyukuk terrane appears to be inconsistent with the general continuity of Late Cretaceous sedimentary basins across Arctic Alaska and the Canada Basin. We were unable to corroborate the result of *Harris et al.* [1987] because the Albian turbidites in our study area do not preserve the original depositional remanent magnetization. In Arctic studies, paleolatitude anomalies as small as 15° are near the limits of resolution of the paleomagnetic method. Therefore the paleolatitude anomalies that were found in our study and in the study of *Harris et al.* [1987] should be interpreted with caution.

The Neocomian volcanic rocks acquired an overprint of both normal and reversed polarity after folding. By solving for the intersection of remagnetization circles (not corrected for tilt of the bedding), we determined the magnetic direction of the overprint (inclination = 72.1°, declination = 3.3°) with reasonable confidence. The 95% confidence oval encloses the present axial dipole direction and cratonic reference directions possibly as old as Eocene. We can rule out Brunhes age (< 700,000 years) remagnetization because of the dual polarity of the overprint, and the overprint is significantly different from the Late Cretaceous reference direction. From the relatively high unblocking temperature (> 350°C) of the overprint, we infer that it was acquired during a major thermal event in the region, probably associated with early Tertiary volcanism.

Geological considerations and several paleomagnetic results from Cretaceous rocks are consistent with the interpretation that large-scale movements have occurred between the continental blocks of Arctic Alaska and the northeast USSR. The Arctic Alaska microcontinent, which consists of the Brooks Range, North Slope, and Chukotka (USSR), apparently rotated counterclockwise away from the Sverdrup margin as the Canada Basin formed in Early Cretaceous time [*Halgedahl and Jarrard*, 1987]. During the onset of the rotation the Koyukuk terrane was probably not connected to the Arctic Alaska microcontinent because there was not enough room within the Canada Basin to accommodate the paleolatitude of the Koyukuk terrane. The gap between the Arctic Alaska microcontinent and the Koyukuk terrane was perhaps a wide ocean basin that was closed by subduction, the suture now being marked by ophiolites of the Angayucham terrane. The possible suture extends westward into the Anyuy belt of the northeast USSR, where Chukotka collided with the Omolon continental block. In addition to early subduction of oceanic crust, closure of the gap was aided by thrusting in the Brooks Range and perhaps by obduction of the Koyukuk terrane onto the leading edge of Arctic Alaska.

The suturing of the Koyukuk terrane to the Arctic Alaska microcontinent formed the core for subsequent accretion of the southern Alaska terranes. During the Cretaceous and early Tertiary, the terranes of southern Alaska were accreting to the core of Alaska via translation along the Tintina and Denali fault zones and by offscraping of buoyant crust from the Pacific oceanic plates [*Stone et al.*, 1982; *Hillhouse*, 1987]. Paleomagnetic evidence from southern Alaska indicates that by early Eocene time the various terranes of Alaska, with the possible exception of the Prince William terrane, had as-

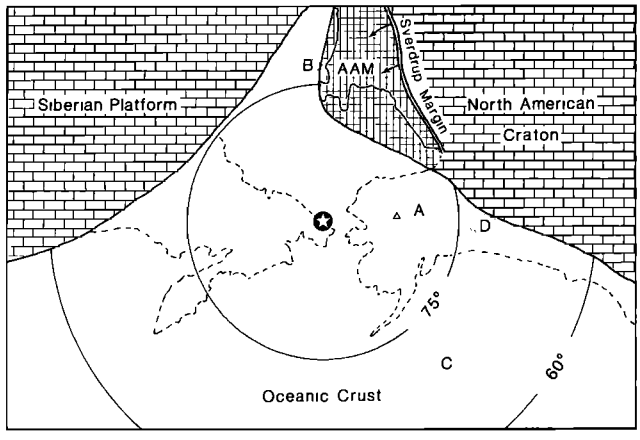


Fig. 14 Hypothetical plate reconstruction of the western Arctic region during Early Cretaceous (Neocomian) time. Arctic Alaska microcontinent (AAM) shown in its position prior to rifting along the Sverdrup margin and opening of the Canada Basin (Figure 1). Paleolatitude range obtained from paleomagnetism of the Koyukuk terrane is between 60° and 75° and is drawn about the Cretaceous reference pole (circled star) from cratonic North America [*Gromme*, 1984]. A–D refer to hypothetical paleopositions of the Koyukuk terrane; see text. The dashed line indicates present coastline of terranes that accreted after Neocomian time. The present position of our study area with respect to cratonic North America is indicated by a triangle.

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sembled together [*Hillhouse et al.*, 1985]. Adjustments in the positions of Arctic terranes occurred throughout the Cenozoic, as indicated by counterclockwise rotation of paleomagnetic poles (68–55 Ma) in southwestern Alaska [*Thrupp and Coe*, 1986; *Coe et al.*, 1985]. These movements were probably the result of deformation that accompanied seafloor spreading in the eastern Arctic Ocean beginning in the early Tertiary [*Harbert et al.*, 1987].

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REFERENCES

Bingham, C., An antipodally symmetric distribution on the sphere, *Ann. Stat.*, *2*, 1201–1225, 1974.

Box, S. E., Early Cretaceous orogenic belt in northwestern Alaska: Internal organization, lateral extent, and tectonic interpretation, in *Tectonostratigraphic Terranes of the Circum-Pacific Region*, edited by D. G. Howell, pp. 137–145, *Earth Sci. Ser.*, vol. 1, Circum-Pacific Council for Energy and Mineral Resources, Houston, Tex., 1985.

Box, S. E., and W. W. Patton, Jr., Volcanic geochemistry of the Koyukuk terrane, western Alaska: Constraints on the origin, evolution, and ultimate collision of an exotic island arc terrane, *J. Geophys. Res.*, in press, 1988.

Box, S. E., W. W. Patton, Jr., and C. Carlson, Early Cretaceous evolution of the northeastern Yukon-Koyukuk basin, west-central Alaska, *U.S. Geol. Surv. Circ.*, *967*, 21–24, 1985.

Bryan, P., and R. G. Gordon, Rotation of the Colorado Plateau: An analysis of paleomagnetic data, *Tectonics*, *5*, 661–667, 1986.

Carey, S. W., The orocline concept in geotectonics, *Proc. R. Soc. Tasmania*, *89*, 255–288, 1955.

Churkin, M., Jr., and J. H. Trexler, Jr., Circum-Arctic plate accretion—Isolating part of a Pacific plate to form the nucleus of the Arctic basin, *Earth Planet. Sci. Lett.*, *48*, 356–362, 1980.

Churkin, M., Jr., J. W. Whitney, and J. F. Rogers, The North American-Siberian connection, a mosaic of craton fragments in a matrix of oceanic terranes, in *Tectonostratigraphic Terranes of the Circum-Pacific Region*, edited by D. G. Howell, pp. 79–84, *Earth Sci. Ser.*, vol. 1, Circum-Pacific Council for Energy and Mineral Resources, Houston, Tex., 1985.

Coe, R. S., B. R. Globerman, P. W. Plumley, and G. A. Thrupp, Paleomagnetic results from Alaska and their tectonic implications, in *Tectonostratigraphic Terranes of the Circum-Pacific Region*, *Earth Sci. Ser.*, vol. 1, edited by D. G. Howell, pp. 85–108, Circum-Pacific Council for Energy and Mineral Resources, Houston, Tex., 1985.

Dalrymple, G. B., Critical tables for conversion of K-Ar ages from old to new constants, *Geology*, *7*, 558–560, 1979.

Diehl, J. F., M. E. Beck, Jr., S. Beske-Diehl, D. Jacobson, and B. C. Hearn, Jr., Paleomagnetism of the Late Cretaceous-early Tertiary north-central Montana alkalic province, *J. Geophys. Res.*, *88*, 10,593–10,609, 1983.

Engebretson, D. C., A. Cox, and R. G. Gordon, Relative motions between oceanic and continental plates in the Pacific basin, *Spec. Pap. Geol. Soc. Am.*, *206*, 59 pp., 1985.

Fisher, R A., Dispersion on a sphere, *Proc. R. Soc. London, Ser. A*, *217*, 295–305, 1953.

Fujita, K., Pre-Cenozoic tectonic evolution of northeast Siberia, *J. Geol.*, *86*, 159–172, 1978.

Fujita, K., and J. T. Newberry, Tectonic evolution of northeastern Siberia and adjacent regions, *Tectonophysics*, *89*, 337–357, 1982.

Globerman, B. R., R. S. Coe, J. M. Hoare, and J. Decker, Palaeomagnetism of lower Cretaceous tuffs from Yukon-Kuskokwim delta region, western Alaska, *Nature*, *305*, 516–520, 1983.

Grantz, A., Strike-slip faults in Alaska, *U.S. Geol. Surv. Open File Rep.*, *267*, 82 pp., 1966.

Grantz, A., S. Eittreim, and D. A. Dinter, Geology and tectonic development of the continental margin north of Alaska, *Tectonophysics*, *59*, 263–291, 1979

Grantz, A., S. Eittreim, and O. T. Whitney, Geology and physiography of the continental margin north of Alaska and implications for the origin of the Canada Basin, in *The Ocean Basins and Margins*, vol. 5, edited by A. E. M. Nairn et al., pp. 439–492, Plenum, New York, 1981.

Gromme, S., Paleomagnetism of Franciscan basalt, Marin County, California, revisited, in *Franciscan Geology of Northern California*, edited by M. C. Blake, Jr., pp. 113–119, Society of Economic Paleontologists and Mineralogists, Pacific Section, Bakersfield, Calif., 1984.

Halgedahl, S., and R. Jarrard, Paleomagnetism of the Kuparuk River Formation from oriented drill core: Evidence for rotation of the Arctic Alaska plate, in *Alaska North Slope Geology*, vol. 2, edited by I. Tailleur and P. Weiner, pp. 581–617, Society of Economic Paleontologists and Mineralogists, Pacific Section, Bakersfield, Calif., 1987.

Halls, H. C., A least-squares method to find a remanence direction from converging remagnetization circles, *Geophys. J. R. Astron. Soc.*, *45*, 297–304, 1976.

Hamilton, W., The Uralides and the motion of the Russian and Siberian platforms, *Geol. Soc. Am. Bull.*, *81*, 2553–2576, 1970.

Harbert, W., L. S. Frei, A. Cox, and D. C. Engebretson, Relative motions between Eurasia and North America in the Bering Sea region, *Tectonophysics*, *134*, 239–261, 1987.

Harland, W. B., A. V. Cox, P. G. Llewellyn, C. A. G. Pickton, A. G. Smith, and R. Walters, *A Geologic Time Scale*, 131 pp., Cambridge University Press, New York, 1982.

Harris, R. A., D. B. Stone, and D. L. Turner, Tectonic implications of paleomagnetic and geochronologic data from the Yukon-Koyukuk province, Alaska, *Geol. Soc. Am. Bull.*, *99*, 362–375, 1987.

Herron, E. M., J. F. Dewey, and W. C. Pitman III, Plate tectonics model for the evolution of the Arctic, *Geology*, *2*, 377–380, 1974.

Hillhouse, J. W., Accretion of southern Alaska, *Tectonophysics*, *139*, 107–122, 1987.

Hillhouse, J. W., and C. S. Gromme, Paleomagnetic studies and the hypothetical rotation of Arctic Alaska, *J. Alaska Geol. Soc.*, *2*, 27–39, 1983.

Hillhouse, J. W., and C. S. Gromme, Early Cretaceous paleolatitude of the Yukon-Koyukuk basin (abstract), *Eos Trans. AGU*, *66*, 1103, 1985.

Hillhouse, J. W., C. S. Gromme, and B. Csejtey, Jr., Tectonic implications of paleomagnetic poles from lower Tertiary volcanic rocks, south central Alaska, *J. Geophys. Res.*, *90*, 12,523–12,535, 1985.

Howell, D. G., D. L. Jones, and E. R. Schermer, Tectonostratigraphic terranes of the circum-Pacific region, in *Tectonostratigraphic Terranes of the Circum-Pacific Region*, edited by D. G. Howell, pp. 3–30, *Earth Sci. Ser.*, vol. 1, Circum-Pacific Council for Energy and Mineral Resources, Houston, Tex., 1985.

Jones, D. L., N. J. Silberling, P. J. Coney, and G. Plafker, Lithotectonic terrane map of Alaska (west of the 141st meridian), scale 1 : 2,500,000, *U.S. Geol. Surv. Misc Field Stud. Map.*, *MF-1874A*, 1987.

Kirschvink, J. L., The least-squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. Astron. Soc.*, *62*, 699–718, 1980.

Lawver, L. A., and A. Baggeroer, A note on the age of the Canada Basin, *J. Alaska Geol. Soc.*, *2*, 57–66, 1983.

Mardia, K. V., and P. J. Zemroch, Table of maximum likelihood estimates for the Bingham distribution, *J. Stat. Comput. Simulations*, *6*, 29–34, 1977.

May, S R., and R. F. Butler, North American Jurassic apparent polar wander: Implications for plate motion, paleogeography, and Cordilleran tectonics, *J. Geophys. Res.*, *91*, 11,519–11,544, 1986.

Miller, T. P., and M. A. Lanphere, K-Ar age measurements on obsidian from the Little Indian River locality in interior Alaska, *U.S. Geol. Surv. Circ.*, *823-B*, B39–B42, 1981.

Onstott, T. C., Application of the Bingham distribution function in paleomagnetic studies, *J. Geophys. Res.*, *85*, 1500–1510, 1980.

Parfenov, L. M., and B. A. Natal'in, Mesozoic evolution of northeastern Asia, *Tectonophysics*, *127*, 291–304, 1986.

Patton, W. W., Jr., Reconnaissance geology of the northern Yukon-Koyukuk Province, Alaska, *U.S. Geol. Surv. Prof. Pap.*, *774-A*, A1–A17, 1973.

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Patton, W. W., Jr., and T. P. Miller, Regional geologic map of the Hughes quadrangle, Alaska, scale 1 : 250,000, *U.S. Geol. Surv. Misc. Geol. Invest. Map, I-459*, 1966.

Patton, W. W., Jr., T. P. Miller, R. M. Chapman, and W. Yeend, Geologic map of the Melozitna quadrangle, Alaska, scale 1 : 250,000, *U.S. Geol. Surv. Misc. Invest. Ser. Map, I-1071*, 1978.

Schmidt, P. W., Bias in converging great circle methods, *Earth Planet. Sci. Lett.*, *72*, 427–432, 1985.

Seslavinskiy, K. B., The South Anyuy geosuture, western Chukotka, *Dokl. Akad. Nauk SSSR*, *249*, 1181–1185, 1979. (*Dokl. Acad. Sci. USSR Earth Sci. Sect.*, Engl. Transl., *249*, 78–81, 1982.)

Steiner, M. B., and C. E. Helsley, Reversal pattern and apparent polar wander for the Late Jurassic, *Geol. Soc. Am. Bull.*, *36*, 1537–1543, 1975.

Stone, D. B., B. C. Panuska, and D. R. Packer, Paleolatitudes versus time for southern Alaska, *J. Geophys. Res.*, *87*, 3697–3707, 1982.

Sweeney, J. F., E. Irving, and J. W. Geuer, Evolution of the Arctic basin, *Publ. Earth Phys. Branch Can.*, *45*(4), 91–100, 1978.

Tailleur, I. L., Rifting speculation of the geology of Alaska's North Slope, *Oil Gas J.*, *67*(39), 128–130, 1969.

Taylor, P. T., L. C. Kovacs, P. R. Vogt, and G. L. Johnson, Detailed aeromagnetic investigation of the Arctic Basin, 2, *J. Geophys. Res.*, *86*, 6323–6333, 1981.

Thrupp, G. A., and R. S. Coe, Early Tertiary paleomagnetic evidence and the displacement of southern Alaska, *Geology*, *14*, 213–217, 1986.

Til'man, S. M., A. N. Afitskiy, and A. D. Chekhov, Comparative tectonics of the Alazey and Oloy zones (northeast USSR) and the problem of the Kolyma Massif, *Geotectonics*, *11*(4), 245–251, 1977.

Vogt, P. R., P. T. Taylor, L. C. Kovacs, and G. L. Johnson, The Canada Basin: Aeromagnetic constraints on structure and evolution, *Tectonophysics*, *89*, 295–336, 1982.

Witte, W. L., D. B. Stone, and C. G. Mull, Paleomagnetism, paleobotany, and paleogeography of the Cretaceous North Slope, Alaska, in *Alaska North Slope Geology*, vol. 2, edited by I. Tailleur and P. Weiner, pp. 571–579, Society of Economic Paleontogists and Mineralogists, Pacific Section, Bakersfield, Calif., 1987.

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